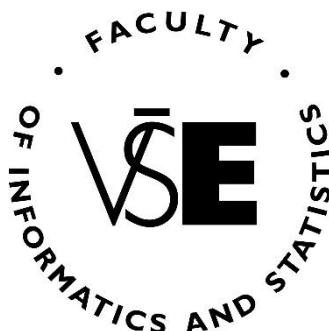


Prague University of Economics and Business

Faculty of Informatics and Statistics



**STAY OR LEAVE: AN ANALYSIS OF DECISION-
MAKING FACTORS AMONG INTERNATIONAL
STUDENTS IN THE CZECH REPUBLIC**

BACHELOR THESIS

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Abstract

This thesis analyses decision-making factors of international student post-study stay or departure in the Czech Republic. Drawing on survey data from international degree-seeking students, provided by the Czech National Agency for International Education and Research (DZS), it uses binary logistic regression to investigate how student experience factors and demographic variables relate to this intention. Key findings indicate that the strongest predictors are the student's country of origin and their work intensity during studies. High academic satisfaction, positive social integration, and satisfaction with pre-arrival bureaucratic procedures also emerged as significant factors positively associated with the intention to stay. The thesis has implications for Czech higher education institutions and policymakers in developing conditions for the post-study stay of international graduates, benefiting both the graduates and the host nation.

Keywords

International Students, Czech Republic, Higher Education, Post-Graduation Intentions, Logistic Regression, Work Intensity, Country of Origin, Academic Satisfaction, Social Integration, Bureaucracy.

Contents

A List of Figures.....	5
Introduction	6
1 Theoretical basis	7
1.1 Theoretical Framework of Student Retention	7
1.1.1 The Push-Pull Framework: Macro-Level Determinants	7
1.1.2 Institutional Barriers: Bureaucracy as Gatekeeping.....	8
1.1.3 Social Integration: Social Capital and Acculturation.....	8
1.1.4 Academic Environment and Institutional Support	9
1.1.5 Human Capital Theory.....	9
1.2 Methodological Basis: Binary Logistic Regression	10
1.2.1 Justification for Logistic Regression and the Logit Model	10
1.2.2 Interpretation using Odds Ratios	10
1.2.3 Model Adequacy Assessment	10
2 Methodology.....	12
2.1 Research Design and Data	12
2.1.1 Sample and Variable Construction	12
2.1.2 Dependent Variable (Stay)	13
2.1.3 Independent Variables (Predictors)	13
2.1.4 Variable Transformation	18
2.1.5 Ethics	19
2.2 Data Analysis	19
2.2.1 Exploratory Data Analysis (EDA).....	19
2.2.2 Assumption Checks	20
2.3 Regression Model	20
2.3.1 Model Specification	20
2.3.2 Model Evaluation	21
2.3.3 Limitations	21
3 Practical Analysis.....	22
3.1 Descriptive Statistics of the Analytic Sample and EDA	22
3.1.1 Dependent Variable.....	24
3.1.2 Independent Continuous Factors	24
3.1.3 Independent Categorical Variables	30
3.1.4 Note on Variable Selection: Exclusion of Gender	33

3.1.5 Multicollinearity check	35
3.2 Results of the Logistic Regression Model	36
3.2.1 Interpretation of Significant Predictors	39
3.2.2 Model Quality and Reliability	42
4 Discussion of Key Findings	47
4.1 Country of Origin	47
4.2 Work Intensity	48
4.3 Academic and Social Factors	48
4.4 Bureaucratic Experience.....	49
Conclusion	50
List of sources	51

A List of Figures

Figure 3.1 Distribution of Students by Intention to Stay in the Czech Republic	24
Figure 3.2 Distribution of Composite Factor Scores	25
Figure 3.3 Factor Comparison for Students Planning to Leave vs. Stay	26
Figure 3.4 Proportion of Students Wishing to Stay by Tuition Satisfaction	27
Figure 3.5 Proportion of Students Wishing to Stay by Accommodation Satisfaction	28
Figure 3.6 Proportion of Students Wishing to Stay by Transport Satisfaction	29
Figure 3.7 Proportion of Students Wishing to Stay by Satisfaction with Other Costs.....	30
Figure 3.8 Proportion of Students Wishing to Stay by Level of Study	31
Figure 3.9 Proportion of Students Wishing to Stay by Country of Origin	32
Figure 3.10 Proportion of Students Wishing to Stay by Work Intensity.....	33
Figure 3.11 Proportion of Students Wishing to Stay by Gender	34
Figure 3.12 Correlation Matrix Between Factors	35
Figure 3.13 ROC Curve	42
Figure 3.14 Calibration Plot	44

Introduction

University students worldwide have increased their participation in studying abroad over the last years. Many young people now choose to study abroad to gain international academic and life experience. Not just the students themselves are influenced by this trend, but also the institutions and countries that host them. The Czech Republic shows steady international student development, and the most recent data shows a significant international student population within its universities (DZS, 2024).

The large number of international students brings several real benefits. Economically, they contribute significantly to the country's economy. Academically and culturally, they diversify the university environment, fostering a dynamic study atmosphere that benefits both domestic and international students. Since their numbers are continuing to grow, it becomes increasingly important to understand their intentions after graduation. International students, in many cases, have the option of staying in the Czech Republic, going back to their country of origin, or moving to a third country. These decisions have a significant impact on the Czech Republic's labor market, education policy, and the retention of skilled graduates (DZS, 2024).

This thesis investigates the specific factors that shape the post-graduation intentions of international students in the Czech Republic. The main research question is: What factors are associated with an international student's intention to stay in the Czech Republic after graduation? To answer this, the thesis uses quantitative methods to analyze cross-sectional survey data from 4,169 degree-seeking students, provided by the Czech National Agency for International Education and Research (DZS). Specifically, it employs binary logistic regression to model how academic satisfaction, social integration, bureaucratic experiences, work intensity, and country of origin relate to the binary outcome of “Stay” or “Leave”.

The thesis is structured into three main parts. Chapter 1, Theoretical Basis, provides the methodological foundation for the analysis. It explains the limitations of Ordinary Least Squares (OLS) for binary data and details the theoretical principles of the chosen method, logistic regression, from its core concepts of log-odds to the interpretation of odds ratios and key model adequacy metrics. Chapter 2, Methodology, describes the research design, the DZS data, and the process of variable construction, including the creation of five composite factors. It also outlines the data analysis workflow, assumption checks, and model specification. Chapter 3, Practical Analysis, presents the results, descriptive statistics and the interpretation of the final logistic regression model, an assessment of its key predictors and an evaluation of its quality and reliability.

1 Theoretical basis

1.1 Theoretical Framework of Student Retention

This chapter establishes the theoretical and methodological foundation for the thesis. First, it examines the key theoretical determinants of student retention to justify the selection of variables such as Country of Origin, Academic Satisfaction, and Social Integration, as well as Institutional Services, Bureaucracy, and Economic determinants. Second, it describes the mathematical principles of logistic regression which is the statistical method used to analyze these factors.

1.1.1 The Push-Pull Framework: Macro-Level Determinants

To understand why international students might stay or return home, Lee's (1966) theory of migration provides a relevant framework. Lee defines migration as a comparison between factors associated with the area of origin and factors associated with the area of destination. Push factors correspond to negative conditions at the origin drive people away (p. 50). These range from economic stagnation (pp. 48, 53) and dissatisfaction with facilities like school systems (p. 50) to oppressive laws (p. 48) and injustice (p. 52).

In this study the variable Country of Origin represents the balance of push and pull factors relative to the student's home country. While push factors vary by origin, the pull factors associated with the destination, such as academic quality, social integration, and economic opportunities, are examined in the following sections through specific student experience variables. Assuming that the objective pull factors of the Czech Republic are constant for all respondents, significant differences in intentions to stay across nationalities likely reflect the varying strength of push factors such as economic instability or political restrictions in their countries of origin.

Beyond these general push factors, the specific conditions of the student's origin country also influence the likelihood of return migration through other channels. Return migration is often driven by acquiring skills abroad that are more valuable in the home country (Dustmann & Görlach, 2016, pp. 102, 115). Non-economic factors are equally crucial as many migrants prefer their home culture (Dustmann & Görlach, 2016, p. 111) or return to be with their families (Dustmann & Görlach, 2016, p. 119). Conversely, strict visa rules may force migrants to leave earlier than planned (Dustmann & Görlach, 2016, p. 121).

Finally, the decision to move involves a third element known as intervening obstacles (Lee, 1966, p. 51). These barriers act as a selective filter or "friction" in the movement process. In the context of this study, these obstacles are seen as institutional and regulatory barriers, which are defined and analyzed in the following section.

1.1.2 Institutional Barriers: Bureaucracy as Gatekeeping

To define the Bureaucracy Factor, this study uses the concept of "intervening obstacles" from Lee's (1966) Push-Pull model. Lee explained that migration always involves barriers; even with strong reasons to move, these obstacles can stop the process or make it too difficult (Lee, 1966, pp. 50–51). For international students, these obstacles are mostly institutional rules and regulations.

Modern migration theory views regulations, like visa policies, as tools to control who enters the country. These rules act as filters that determine whether a student can become a resident. While they may seem like routine paperwork, complex requirements create real obstacles to migration. As Dustmann and Görlach (2016) note, strict laws often force migrants to leave early, even if they have strong personal or economic reasons to stay (p. 120).

This study emphasizes the distinction between the official rules and the student's subjective perception of these barriers. The Bureaucracy Factor in this study measures the student's personal experience of these obstacles. It combines views on clarity, cost, and helpfulness regarding three key processes: admissions, visa processing, and recognition of prior education. According to the theory, if these processes are seen as unclear, expensive, or unfriendly, the "psychological cost" of migration increases. (Lee, 1966, p. 51).

In contrast, a transparent and supportive bureaucratic environment reduces barriers. Therefore, high satisfaction with administrative processes is likely to increase the intention to stay, as it indicates a supportive environment and makes settling in easier.

1.1.3 Social Integration: Social Capital and Acculturation

This study applies social capital and acculturation theories to examine how relationships affect the decision to remain in a host country. It specifically distinguishes between bonding and bridging social capital. Bonding social capital refers to "inward looking" connections within "homogeneous groups" that strengthen shared identities (Wang, 2020, p. 40). In this study, sense of belonging among other international students measures this concept. While these networks offer support, they link people with similar resources and rarely help spread new information (Wang, 2020, p. 41).

In contrast, bridging social capital connects people who are "different from each other" and is essential for "resource acquisition". Wang (2020) defines this as connections with people outside one's own national group, including locals and other international students. In this study, it is measured by a sense of inclusion among local students or the wider community. Bridging ties are crucial because they connect students to the host society beyond their co-national communities and help them access resources and information unavailable in their bonding networks.

Berry et al.'s (2006) identify integration as the most effective strategy (p. 306). They found that integrated youth have better psychological and sociocultural outcomes than those who separate or assimilate (pp. 320–321). Therefore, successful integration fosters a sense of belonging, which encourages students to stay.

1.1.4 Academic Environment and Institutional Support

This study uses Kahu and Nelson's (2018) "educational interface" framework to justify including Academic and Services Factors. Unlike older theories that viewed student transition as a temporary phase, this framework describes it as an ongoing, active process (Kahu & Nelson, 2018, p. 1). Success is determined within this interface, described as the psychosocial space where university systems and individual student characteristics interact (Kahu & Nelson, 2018, pp. 1, 6).

The Academic Factor reflects the framework's "structural" (curriculum) and "psychosocial" (teaching) influences (Kahu & Nelson, 2018, p. 7). Quality teaching and curriculum design are essential because they activate key mechanisms like self-efficacy and emotions (Kahu & Nelson, 2018, p. 8). When academic tasks match student skills, academic self-efficacy and positive emotional engagement increase (Kahu & Nelson, 2018, p. 8). Conversely, a poor fit in this area leads to anxiety and disengagement (Kahu & Nelson, 2018, p. 9).

The Services Factor relates to institutional policies (Kahu & Nelson, 2018, p. 7) and supports wellbeing, a key element of student success. Supportive services help reduce stress caused by "lifeload," which is vital for non-traditional students (Kahu & Nelson, 2018, p. 10). Moreover, effective environments build a sense of belonging, defined as the student's perceived fit with the institution (Kahu & Nelson, 2018, p. 9).

Although the framework focuses on immediate learning, positive engagement leads to long-term results, such as retention. For students, the university is the primary space where they "live and learn" (Kahu & Nelson, 2018, p. 8). Therefore, promoting a sense of belonging and self-efficacy is expected to encourage the broader decision to stay in the Czech Republic.

1.1.5 Human Capital Theory

To analyze the Work and Value factors, this study applies Human Capital Theory (HCT). This theory views individuals as economic agents who invest in education to "boost their earning potential and overall economic value" (Calonge et al., 2023, p. 3). The Value Factor represents a real-time assessment of this return on investment. Within this framework, if current financial burdens (living costs, tuition) outweigh the perceived quality of the degree, the investment becomes less attractive, leading to a decision to leave the country.

The variable *Work* measures human capital accumulation. Beyond financial resources, working develops "specific human capital" through experience (Calonge et al., 2023, p. 3). This accumulation of "country-specific human capital" (Calonge et al., 2023, p. 9), including local networks and cultural knowledge, enhances competitiveness and employment outcomes. Conversely, a lack of such capital acts as a barrier to retention.

Crucially, work intensity performs in two ways: beyond the economic accumulation of skills, it encourages social integration by building "bridging social capital" (see Section 1.1.3) through professional socialization.

After identifying these determinants, the following section justifies binary logistic regression as the appropriate method for the analysis.

1.2 Methodological Basis: Binary Logistic Regression

Logistic regression is a primary and highly useful method in the social sciences. It is frequently selected for studying binary outcomes, which are variables with only two possible values. These "yes/no" or "either/or" scenarios include examples like voting intention (yes/no), employment status (employed/unemployed), or, as in this thesis, the decision to remain in or leave a country.

1.2.1 Justification for Logistic Regression and the Logit Model

Since the dependent variable is binary (Stay vs. Leave), Ordinary Least Squares (OLS) regression is not appropriate as it may produce predicted probabilities outside the feasible range of 0 to 1 (Pampel, 2021, p. 14). Therefore, this study uses binary logistic regression to model the relationship between predictors and the natural logarithm of the odds.

The model is defined as:

$$\ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

1.2.2 Interpretation using Odds Ratios

Because log-odds coefficients (β) are not intuitively interpretable, results are presented as Odds Ratios (OR). These are calculated by exponentiating the coefficients:

$$OddsRatio(OR) = e^{\beta}$$

An $OR > 1$ suggests that the predictor increases the likelihood of staying, while an $OR < 1$ indicates a decrease (Pampel, 2021, p. 28). An OR of 1 implies no effect. This method clearly shows how variables, such as academic satisfaction or work intensity, influence the odds of retention.

1.2.3 Model Adequacy Assessment

Unlike Ordinary Least Squares regression, where the R^2 value clearly measures explained variance, logistic regression lacks a single standard metric. Therefore, assessing model fit requires looking at several different measures of performance (Dey et al., 2025, p. 4). This multi-faceted approach is necessary because different statistics capture different aspects of model quality.

Assessing Explanatory Power: Pseudo- R^2 Measures

To offer a measure of explanatory power similar to OLS, various Pseudo- R^2 statistics were created. However, widely used measures like Cox & Snell R^2 cannot reach a maximum value of 1, making interpretation difficult (Ünsal Özbek & Yetkiner, 2021, p. 26). To fix this problem, Nagelkerke (1991) suggested an adjusted version. This version adjusts the statistic

so that its maximum possible value is 1 (Ünsal Özbek & Yetkiner, 2021, p. 26). Because the Nagelkerke R^2 provides a full 0 to 1 range, it is often considered a more intuitive measure and is the one used in this thesis to represent the proportion of variation explained.

Assessing Discriminative Ability (AUC-ROC)

A logistic regression model must also be evaluated for its discrimination, which refers to the ability to accurately distinguish between low and high-risk individuals (Crowson et al., 2016, p. 1). This is measured by the Area Under the Receiver Operating Characteristic Curve (AUC).

The AUC represents the probability that a randomly selected positive case ranks higher than a randomly selected negative case (Li, 2024, p. 1). An AUC of 0.5 means the model has no ability to discriminate between groups, whereas a value of 1 indicates perfect discrimination ability (Dey et al., 2025, p. 4). This metric is particularly robust because it remains valid regardless of the baseline class proportions (Li, 2024, p. 11).

Assessing Calibration and the Hosmer-Lemeshow Test

While discrimination checks if a model can rank cases correctly, calibration checks if its predicted probabilities are accurate. A common method to test this goodness of fit is the Hosmer-Lemeshow test (Crowson et al., 2016, p. 2; Řezanková, 2024, p. 171). The null hypothesis of this test is that the model is well calibrated, meaning its predictions fit the observed data well (Řezanková, 2024, p. 171).

However, this test has major limitations. Its results depend on the way the groups are created and it suffers from low statistical power as its resulting p -value is uninformative regarding the nature or extent of any miscalibration (Van Calster et al., 2019, p. 4; Crowson et al., 2016, p. 2). For these reasons, using the Hosmer-Lemeshow test alone is not recommended. A visual check using a calibration curve is preferred, as it clearly shows the relationship between predicted risk and observed outcomes (Van Calster et al., 2019, p. 4).

2 Methodology

2.1 Research Design and Data

In this thesis, I use secondary data from a cross-sectional survey conducted by the Czech National Agency for International Education and Research (DZS) during the 2022/23 academic year.

Between October 2022 and February 2023, the DZS distributed this online questionnaire via all public, state, and private universities in the Czech Republic. The survey, offered in Czech and English, was sent to all full-time, degree-seeking international students (DZS, 2024). Exchange program students were intentionally excluded, as this research focuses on students who might realistically settle in the country long-term after graduation.

The questionnaire covered several thematic areas relevant to the student experience:

- Demographics (gender, nationality, program language, level of study);
- Pre-arrival procedures (admissions, visa process, recognition of prior education);
- Academic experience (program quality, teaching methods, study materials);
- University services and social life (administrative support, facilities, integration);
- Cost and value perceptions (tuition fees, living costs);
- Future plans (intention to stay in or leave the Czech Republic after graduation).

The initial survey produced 6,388 fully completed records.

2.1.1 Sample and Variable Construction

To prepare the raw data for this study's analysis, I applied two filters:

- **Outcome Clarity:** I removed respondents who selected "I do not know" or "Other" for their post-graduation plans. This step was necessary because the logistic regression model requires a clear binary outcome (in this case, 'Stay' or 'Leave'). As Dey et al. (2025, p. 2) note, this model is designed for "binary classification where the response variable is limited to two values and represented as either 0 or 1".
- **Complete Information:** Missing data is a common issue in real-world surveys, and how to handle these gaps is a crucial decision. As Dey et al. (2025, p. 7) state, "it's always important to properly manage missing data, as this will avoid incorrect conclusions." Failing to address this can significantly compromise the study's results. Ignoring missing data is not a viable option. Dey et al. (2025) warn that listwise deletion ("elimination of instances with missing data") "would lead to a reduced sample size together with a loss of vital information," affecting validity (p. 7).
To avoid these biases and preserve the sample size, I rejected listwise deletion in favor of **Single Imputation**.

To address missing values in the predictors, I employed Single Imputation using the Predictive Mean Matching (PMM) algorithm via the mice package in R. As Nguyen (2025) explains, PMM imputes values based on similar observed data points ("donors") rather than calculating mathematical averages. This ensures that imputed values remain realistic integers—crucial for Likert scales—and preserves the original data distribution better than simple mean imputation.

The imputation procedure involved three steps:

- **Filtering:** Records where the dependent variable (Stay) was missing were excluded to ensure the model relies only on actual reported intentions.
- **Formatting:** Categorical variables (Level, Country) were explicitly converted to factors.
- **Execution:** The imputation was generated using the parameters $m = 1$ and `method = 'pmm'` to create the final complete dataset.

Single Imputation was selected for this study to maintain interpretability and computational simplicity. As Nguyen (2025) notes, Single Imputation provides a complete dataset that can be analyzed using standard techniques without the complexity of pooling results from multiple datasets.

After these steps, the final analytic sample contains 4,225 students.

2.1.2 Dependent Variable (Stay)

The main outcome for this study is the student's intention to stay in the Czech Republic after their studies. I created a binary variable named Stay based on the survey question, "What are your plans after graduation?".

- **Stay = 1 (Plan to Stay):** This code was assigned if the student chose "I plan to continue my studies in the Czech Republic" or "I plan to work in the Czech Republic."
- **Stay = 0 (Plan to Leave):** This code was assigned for all other definitive plans, such as returning to their home country or moving to a different country for work or study.

2.1.3 Independent Variables (Predictors)

To create meaningful predictors, I grouped individual questions into composite factors. Survey questionnaires do not always measure complex ideas directly, and measuring abstract concepts like "social integration" with a single question is unreliable due to potential measurement error (Evans et al., 2022, p. 1429). To address this, I followed the standard sociological practice of creating composite factors (indices). As noted by Evans et al. (2022), combining related items into an index captures complex concepts more reliably than single items. Similarly, Wu et al. (2025) employed this approach by reverse-coding and combining items to measure mastery. Following this logic, I constructed five composite factors by averaging the student's scores on related questions.

I initially created five composite factors by averaging the student's scores on related questions. To ensure the validity of these indices, I assessed their internal consistency using Cronbach's Alpha. Based on the analysis, four factors (Academic, Social, Services, and Bureaucracy) demonstrated good reliability ($\alpha > 0.8$) and were retained as composite measures.

However, the Value factor showed poor internal consistency ($\alpha = 0.586$). The correlation matrix revealed weak associations between items (0.15 to 0.42), suggesting they lack the homogeneity needed for a single construct. Izah et al. (2024, p. 8) note that Cronbach's Alpha "may underestimate reliability if there is insufficient correlation between items". Consequently, I analyzed the economic aspects as four separate single-item variables rather than a composite factor.

Composite Factors: For each of the four factors, I created a single continuous variable by averaging the student's scores on the related questions. All items used a 4-point scale, where a higher score means a more positive perception.

- **Academic Factor:** This is an average of 7 questions about satisfaction with the study program, including its quality, connection to practice, materials, and teacher performance. The factor is composed of the following items:
 - Do you consider your degree program to be of high quality?
 - The level of English of the teachers in my degree program is sufficient.
 - My degree program is sufficiently connected to real life practice.
 - The teachers sufficiently reflect my needs as an international student.
 - My degree program offers study materials of sufficient quality.
 - I consider the classroom/lab equipment sufficient.
 - Please rate the proportion of quality teachers in your study program.
- **Social Factor:** This is an average of 6 questions related to social integration, such as feeling included, integration into university life, and the role of student organizations. The factor is composed of the following items:
 - Student organisations or clubs
 - The student inclusion system (e.g. a Buddy program, adaptation courses, etc.)
 - I feel included among Czech students.
 - I feel included among other international students.
 - I feel integrated into the everyday life of my university.
 - I find the local community welcoming to the needs of international students.
- **Services Factor:** This is an average of 8 questions assessing the quality of university services, including administrative staff, IT systems, facilities, and counselling. The factor is composed of the following items:
 - Do you consider the services for international students at your university to be of sufficient quality?

- The attitude of the administrative staff
 - The university's IT system
 - The availability and clarity of information
 - The university's facilities (e.g. library, canteen, etc.)
 - Accommodation in dormitories
 - The selection of activities provided by the university beyond the regular curriculum
 - Counselling services (e.g. career, psychological, etc.)
- **Bureaucracy Factor:** This is an average of 12 questions assessing administrative barriers faced before arrival, such as the clarity and cost of the admissions, visa, and education recognition processes. The factor is composed of the following items:
 - I consider the admissions procedure and the requirements for admission to be clear.
 - The difficulty of the entrance exam was adequate.
 - The fees associated with the admissions procedure were low.
 - The admissions procedure was not time-consuming/administratively demanding.
 - I find the visa requirements and the administrative process of visa processing to be clear.
 - The consular services were available.
 - The costs associated with the visa process were low.
 - The visa application process was not time-consuming/administratively demanding.
 - I find the requirements for recognition process clear and understandable.
 - The approach of stakeholders in the recognition process was helpful.
 - The fees associated with the recognition process were low.
 - The recognition process was not time consuming / administratively demanding.

These factors were calculated by taking the row-wise mean of the relevant question columns for each student. In the R software, I used the mutate function to create new columns for each factor and the rowMeans function to calculate this average score.

- **Financial Variables:** Because the combined scale showed low consistency, the following items were analyzed separately:
 - Tuition: "How do you rate the quality of your degree program compared to the level of tuition fees?"
 - Accommodation: Satisfaction with accommodation costs.
 - Transport: Satisfaction with transport costs.
 - Other Costs: Satisfaction with other living costs.

Code 2.1

Calculating Composite Factor Variables (Source: Author's own processing using R)

```
df_factors <- df_clean %>%
  mutate(
    Academic = rowMeans(select(., all_of(academic_cols)), na.rm = TRUE),
    Social    = rowMeans(select(., all_of(social_cols)), na.rm = TRUE),
    Services  = rowMeans(select(., all_of(services_cols)), na.rm = TRUE),
    Bureaucracy = rowMeans(select(., all_of(bureaucracy_cols)), na.rm = TRUE),
    Tuition = .[[67]],
    Accommodation = .[[107]],
    Transport = .[[108]],
    Other_Costs = . [[109]],
    Work = .[[118]]
  )
```

Other predictors:

- **Work:** A variable measuring work intensity during studies, recoded as a 5-point scale (1 = Not working, 5 = More than 20 hours/week).
- **Level:** The student's current level of study (Bachelor's, Master's, or Doctoral).
- **Country:** The student's country of origin. To ensure the groups were large enough for analysis, I created this variable by identifying the top 10 most represented nationalities in the data. All other nationalities were combined into a single "Other" category, which is used as the reference level in the regression model.

The survey also collected the respondent's **Gender**. I initially considered using it in the model but first needed to check if it was statistically significant and if it reduced the sample size. This analysis is explained in Chapter 3.

I used R code to prepare this data. First, the code identified the top 10 most frequent countries. Then, it created the new Country and Level variables. After this, I created the dependent variable Stay, as described in section 2.1.2.

Finally, the code selected only the columns needed for the model, converted categorical variables into factors for the regression, and set "Other" as the baseline reference category. To address missing values without reducing the sample size, I applied Single Imputation using the PMM algorithm, which produced the final complete analytic sample.

First, to ensure statistical validity, I grouped the nationalities. Using the `top_n` function, I identified the 10 most frequent countries of origin in the sample. All other nationalities were recoded into a single "Other" category to serve as a robust baseline for the regression.

Code 2.4

Demographic Variables (Source: Author's own processing using R)

```
countries <- df_labeled %>%
  count(.data[[q_country]], sort = TRUE) %>%
  top_n(10, n) %>%
  pull(.data[[q_country]])

df_factors$Country <- if_else(
  df_labeled[[q_country]] %in% countries,
  df_labeled[[q_country]],
  "Other"
)
df_factors$Level <- df_labeled[[q_level]]
```

Next, I constructed the binary dependent variable Stay. I also selected the final set of predictors.

Code 2.7

Constructing the Dependent Variable and Selecting Predictors (Source: Author's own processing using R)

```
df_pre_final <- df_factors %>%
  mutate(
    Stay = case_when(
      # Answers 1 ("continue studies") and 2 ("work in CR") -> Stay (1)
      df_factors[[119]] %in% c(1, 2) ~ 1,
      # Answers 3-6 (all "leave" options) -> Leave (0)
      df_factors[[119]] %in% c(3, 4, 5, 6) ~ 0,
      # "I don't know" or "Other" become NA
      TRUE ~ NA_real_
    )
  ) %>%
  select(
    Academic, Social, Services, Bureaucracy,
    Tuition, Accommodation, Transport, Other_Costs,
    Work, Level, Country, Stay
  )
```

Before performing imputation, I removed observations where the outcome variable (Stay) was missing, as the goal is to predict the intention, not to impute it. I also converted the categorical demographic variables into factors and set "Other" as the reference level to ensure correct interpretation of the regression coefficients.

Code**2.10***Data Preparation for Imputation (Source: Author's own processing using R)*

```
df_for_imp <- df_pre_final %>%
  filter(!is.na(Stay))
df_for_imp$Level <- as.factor(df_for_imp$Level)
df_for_imp$Country <- as.factor(df_for_imp$Country)
df_for_imp$Country <- relevel(df_for_imp$Country, ref = "Other")
```

Finally, I used the mice package to impute missing values in the predictor variables. I used the Predictive Mean Matching (pmm) method with a single iteration ($m = 1$) to generate the final complete dataset for analysis.

Code**2.13***Executing Single Imputation (Source: Author's own processing using R)*

```
imp_obj <- mice(df_for_imp, m = 1, method = 'pmm', seed = 123, print = FALSE)
df_final <- complete(imp_obj)
```

Code 2.14

```
Executing Single Imputationimp_obj <- mice(df_for_imp, m = 1, method = 'pmm', seed = 123,
print = FALSE)
df_final <- complete(imp_obj)
```

2.1.4 Variable Transformation

To ensure consistency across the dataset, I reversed several items that were originally coded on an inverse scale (e.g., a score of 1 became 4, 2 became 3, etc.). This step ensures that for all factor items, a higher value consistently represents a more positive or favorable view.

Additionally, I transformed one item with a 10-point scale to align with the 4-point scale used for the other factor components.

Code 2.15*Reverse-coding for specific columns (Source: Author's own processing using R)*

```
rev_cols <- c(44:47, 49:56, 61:67, 93:109)

df_clean <- df_clean %>%
  mutate(
    across(all_of(rev_cols), ~ 5 - .),
    across(all_of(118), ~ 6 - .),
    across(all_of(90), ~ 1 + (. - 1) * 4 / 9)
  )
```

I achieved this with the following transformations. The R code first identifies the columns (rev_cols) that needed to be reverse-coded. Using the mutate and across functions, the code applies a simple mathematical operation ($5 - \text{value}$) to invert the 4-point scales. Similar transformations were applied to the work intensity question and the 10-point scale question to standardize them with the rest of the data.

2.1.5 Ethics

Participation in the DZS survey was entirely voluntary and anonymous (DZS, 2024). Before releasing the data, the agency removed all direct identifiers, names, student numbers, IP addresses, and timestamps, ensuring that individual respondents cannot be traced. Each participant clicked an informed consent statement at the start of the questionnaire, satisfying the requirements of the EU General Data Protection Regulation (GDPR) (DZS, 2024). Because the dataset I received contains no personal identifiers and only aggregated variables, no additional ethics approval was required.

2.2 Data Analysis

I performed all data processing and statistical analysis for this thesis in R. Data manipulation and transformation primarily used the dplyr and tidyr packages. All data visualizations were created with ggplot2, and the correlation matrix was generated using the corrplot package. The final logistic regression model was built with the base stats package and its performance was evaluated with metrics from the pROC and pscl packages. The analysis was structured in two main phases: Exploratory Data Analysis (EDA) and modeling.

2.2.1 Exploratory Data Analysis (EDA)

Before building the final model I conducted an exploratory analysis to understand the underlying patterns and distributions in the data. This phase focused on visual and descriptive methods rather than formal hypothesis testing. The key steps included:

- **Analysis of the Dependent Variable:** I examined the distribution of the binary outcome variable Stay to understand the proportion of students planning to stay in the Czech Republic versus those planning to leave. I visualized this using a bar chart to show the count in each category.
- **Analysis of Continuous Predictors:** I assessed the distributions of the four composite factors (Academic, Social, Services, Bureaucracy) using histograms and density plots. To understand their relationship with the outcome variable, I used boxplots to compare the scores of students who planned to stay versus those who planned to leave. I examined the distributions of the four single-item financial variables (Tuition, Accommodation, Transport, Other Costs). I visualized their distributions using bar charts to observe response frequencies.

- **Analysis of Categorical Predictors:** I explored the relationships between categorical predictors (Level, Country, Work) and the decision to stay using normalized, stacked bar charts. This method made it easy to visually compare how the intention to stay varied across different groups (e.g., across different countries or study levels).

2.2.2 Assumption Checks

To ensure the logistic regression model's stability, I checked for multicollinearity (high correlation) among the continuous predictors. This is a crucial step, as the absence of significant multicollinearity is a key assumption of logistic regression (Dey et al., 2025, p. 2). Its presence can lead to unreliable coefficient estimates.

I performed this check by calculating and visualizing a Pearson correlation matrix for the five composite factors and the Work variable, which is a common detection method (Vatcheva et al., 2016, p. 7). The matrix revealed only low to moderate correlations among all predictors. No values were high enough (e.g., typically above 0.80) to suggest a problematic level of multicollinearity that would compromise the model's validity (Vatcheva et al., 2016, p. 7). Therefore, all predictor variables were retained for the final model.

2.3 Regression Model

To analyze the factors influencing a student's decision to stay, I used a binary logistic regression model. This method is appropriate because the dependent variable is dichotomous (1 = Stay, 0 = Leave). The model was estimated using the maximum likelihood method (Řezanková, 2024, p. 170).

2.3.1 Model Specification

Instead of using a phased or exploratory approach, I constructed a single, comprehensive model. This method allows for the simultaneous assessment of all five composite factors (Academic, Social, Services, Value, Bureaucracy) while controlling for the influence of key demographic and situational variables (Work, Level, Country).

For interpretation, the regression coefficients (B) were converted into Odds Ratios (OR) through exponentiation (Řezanková, 2024, p. 166). The Odds Ratio indicates how the odds of a student planning to stay (Stay = 1) change for a one-unit increase in a predictor variable, holding all other variables constant (Řezanková, 2024, pp. 164, 166). An OR greater than 1 signifies a positive influence on the intention to stay, while an OR less than 1 signifies a negative influence.

2.3.2 Model Evaluation

To check the model's performance and reliability, I used three metrics:

- **Nagelkerke Pseudo R^2 :** This metric estimates how much of the variance in the dependent variable (intention to stay) is explained by the predictor variables in my model.
- **Area Under the Curve (AUC):** Derived from the (ROC) curve, the AUC measures how well the model can distinguish between the two outcomes (Stay vs. Leave). An AUC of 0.5 means the model is no better than random chance, while 1.0 means it is a perfect predictor.
- **Calibration Plot:** This plot checks the model's reliability by comparing its predicted probabilities to the actual observed outcomes. In a well-calibrated model, the predictions closely match the real-world results, usually following a 45-degree line.

2.3.3 Limitations

Although the methods used are robust, the findings of this thesis have several limitations:

- **Cross-Sectional Design:** The data was collected at only one point in time. This design cannot establish causality. For example, I cannot know if high academic satisfaction causes an intention to stay, or if students who already plan to stay are just more likely to report high satisfaction.
- **Self-Report Measures:** All data, including both the predictors (satisfaction factors) and the outcome (intention to stay), came from the same self-reported questionnaire. This can sometimes create "common method bias," although using multi-item composite factors can help reduce this risk.
- **Voluntary Participation:** The survey was voluntary. It is possible that students with very strong opinions (either positive or negative) were more likely to respond, which could create a self-selection bias.
- **Sample Representativeness:** The sample was not weighted to fix known imbalances (like the over-representation of Slovak students). Therefore, the findings may not be perfectly generalizable to the entire international student population in the Czech Republic.
- **"Stay" as an Intention:** The model predicts the intention to stay, not the actual behavior of staying. Students' plans might change after graduation due to new opportunities or other unforeseen reasons.

Despite these limitations, the large sample size and the clear approach to building variables and evaluating the model provide a solid empirical basis for this analysis.

3 Practical Analysis

3.1 Descriptive Statistics of the Analytic Sample and EDA

To begin the analysis, I first filtered the data as described in the Methodology chapter. I removed respondents who selected "I do not know" or "Other" for the main outcome variable. To address missing values in the predictors, I employed Single Imputation (as detailed in Section 2.1.1) rather than listwise deletion to preserve the sample size.

This process produced the final analytic dataset `df_final` containing 4,225 complete observations.

I first checked the structure of this dataset using the `str(df_final)` command in R. This command confirmed the total observations (4,225) and variables (12), and verified the data type for each column (as shown in Code 3.1)

Code 3.2

Structure of the Final Dataframe (Source: Author's own processing using R)

```
## 'data.frame': 4225 obs. of 12 variables:
## $ Academic : num 2.94 3.59 3.11 4.14 3.79 ...
## $ Social : num 3.17 3 4 4 3.67 ...
## $ Services : num 2 3.57 3 4 4 ...
## $ Bureaucracy : num 2.5 3.75 3.62 4 3.75 ...
## $ Tuition : num 3 4 4 4 4 4 3 4 4 ...
## $ Accommodation: num 1 4 2 4 4 4 4 2 1 4 ...
## $ Transport : num 2 4 4 4 4 4 4 3 3 4 ...
## $ Other_Costs : num 1 4 2 3 4 4 4 3 3 4 ...
## $ Work : num 2 5 5 5 4 4 2 2 1 4 ...
## $ Level : Factor w/ 3 levels "Bachelor's degree programme",...: 3 3 1 3 1 1 1 3 1 3 ...
## $ Country : Factor w/ 11 levels "Other","Belarus",...: 9 9 9 9 9 9 9 9 9 9 ...
## $ Stay : num 0 1 1 1 1 1 0 0 1 1 ...
```

This check confirmed the data was correctly structured for the following analysis. The four composite factors (Academic, Social, Services, Bureaucracy), the four financial variables (Tuition, Accommodation, Transport, Other Costs), and the Work variable were all correctly stored as numeric (num) types. My key categorical predictors, Level and Country, were correctly set as factors (Factor), which is essential for the regression model to treat them as distinct groups rather than a continuous scale. Finally, the dependent variable, Stay, was correctly stored as a numeric (num) type, representing the binary outcome (0 = Leave, 1 = Stay).

To understand the distribution of each variable, I used the `summary(df_final)` command. This provides a detailed statistical overview:

Code 3.5

Summary Statistics for the Final Dataframe (Source: Author's own processing using R)

```
##      Academic      Social      Services      Bureaucracy
## Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:3.016   1st Qu.:2.500   1st Qu.:2.750   1st Qu.:3.000
## Median :3.426   Median :3.000   Median :3.125   Median :3.364
## Mean   :3.334   Mean   :2.921   Mean   :3.064   Mean   :3.325
## 3rd Qu.:3.794   3rd Qu.:3.400   3rd Qu.:3.500   3rd Qu.:3.667
## Max.   :4.250   Max.   :4.000   Max.   :4.000   Max.   :4.000
##
##      Tuition      Accommodation      Transport      Other_Costs
## Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:3.000   1st Qu.:2.000   1st Qu.:3.000   1st Qu.:2.000
## Median :3.000   Median :3.000   Median :4.000   Median :3.000
## Mean   :3.345   Mean   :2.614   Mean   :3.485   Mean   :2.862
## 3rd Qu.:4.000   3rd Qu.:3.000   3rd Qu.:4.000   3rd Qu.:3.000
## Max.   :4.000   Max.   :4.000   Max.   :4.000   Max.   :4.000
##
##      Work      Level
## Min.   :1.000   Bachelor's degree programme:2240
## 1st Qu.:1.000   Doctoral degree programmes : 633
## Median :2.000   Master's degree programmes :1352
## Mean   :2.799
## 3rd Qu.:4.000
## Max.   :5.000
##
##      Country      Stay
## Other      :1148   Min.   :0.0000
## Slovakia   :1077   1st Qu.:0.0000
## Russian Federation (the): 728   Median :1.0000
## Ukraine    : 492   Mean   :0.5792
## India      : 209   3rd Qu.:1.0000
## Kazakhstan : 187   Max.   :1.0000
## (Other)    : 384
```

The `summary()` output provides the foundation for understanding the analytic sample.

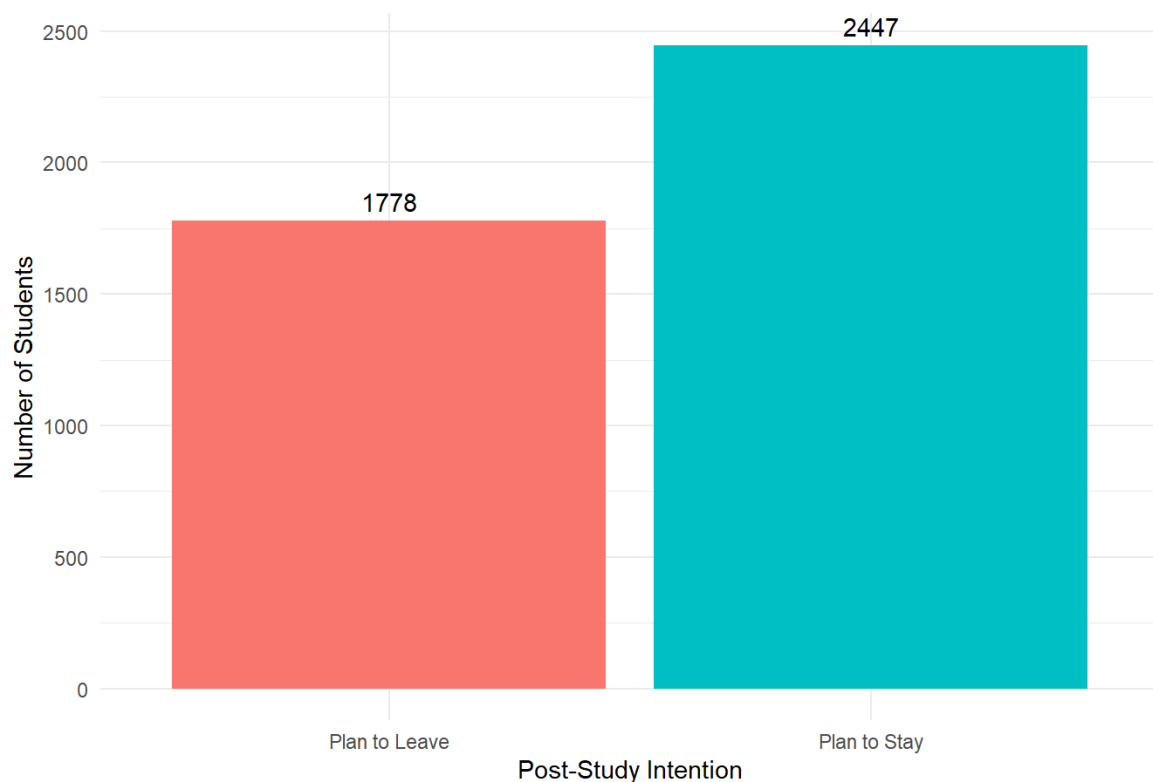
3.1.1 Dependent Variable

I began the analysis by examining the primary outcome variable, Stay. The summary() output (from Code 3.2) shows the mean for Stay is 0.5792. Because 'Stay' was coded as 1 and 'Leave' as 0, this mean value directly represents the proportion of students in the sample who intend to stay.

This indicates that a majority (57.9%) plan to remain in the Czech Republic after their studies. To visualize the absolute numbers, I generated the bar chart in Figure 3.1 . This chart clearly illustrates the distribution: out of 4,225 respondents, 2,447 expressed an intention to stay, while 1,778 plan to leave.

Figure 3.2

Distribution of Students by Intention to Stay in the Czech Republic (Source: Author's own processing using R)



3.1.2 Independent Continuous Factors

Composite factors

Next, I examined the four composite factors I constructed to measure different aspects of the student experience.

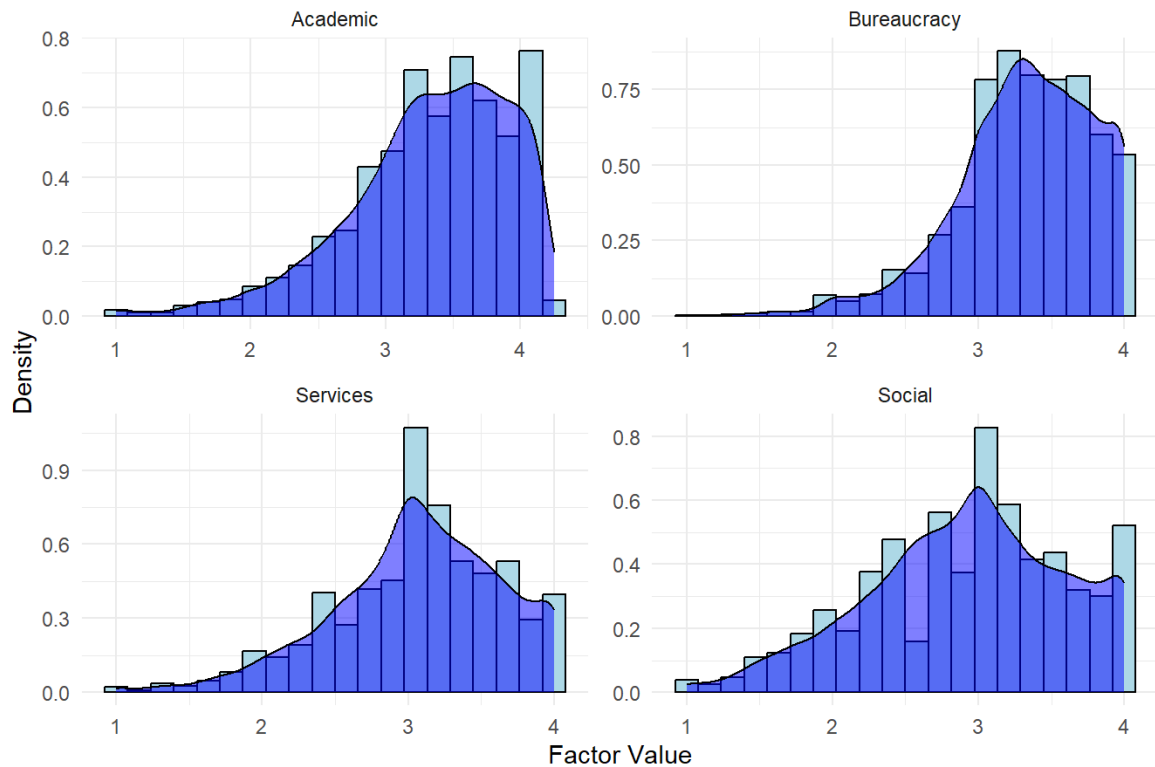
Examining the summary() output, I first noted that the mean values for all four factors were above the theoretical midpoint of 2.5 (on their 1-4 scale). This provides an initial indication that, on average, student perceptions were more positive than negative.

- The Academic factor (M=3.33) and the Bureaucracy factor (M=3.33) had the highest average satisfaction scores.
- The other three factors were also rated positively: Services (M=3.06), and Social (M=2.92).

To understand these factors beyond their averages, I plotted the distribution for each one, as shown in Figure 3.2.

Figure 3.5

Distribution of Composite Factor Scores (Source: Author's own processing using R)



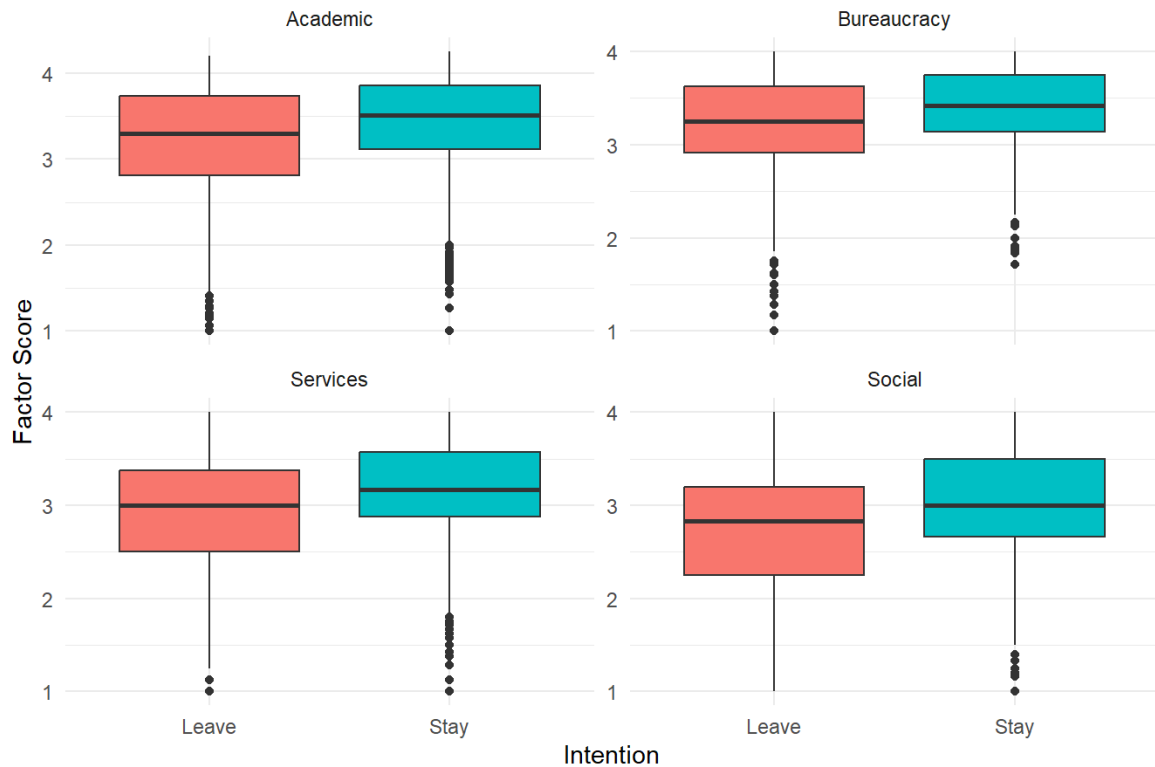
These histograms and density plots confirm what the mean values suggest. All four distributions are negatively skewed, meaning responses are clustered at the positive end of the scale (scores of 3.0 and 4.0). This visually confirms that students in the sample generally reported positive experiences. I also noted from the plots that the Social factor's distribution appears the most balanced, while the Academic and Bureaucracy factors show a strong concentration of high scores.

After analysing the factors individually, the next step was a preliminary bivariate analysis. I wanted to check for a visual link between these factors and the Stay outcome variable. To

do this, I generated the boxplots in Figure 3.3 , which compare the score distributions for students planning to "Leave" versus "Stay".

Figure 3.8

Factor Comparison for Students Planning to Leave vs. Stay (Source: Author's own processing using R)



This visualization revealed a clear and consistent pattern. For all four factors, the median score (the dark horizontal line) is visibly higher for the "Stay" group than for the "Leave" group. This indicates that students who plan to remain reported higher satisfaction across their academic, social, administrative, and service experiences. This visual evidence is the first strong indication that these positive perceptions are associated with a higher likelihood of intending to stay.

It is also important to note the presence of outliers, represented by the black dots below the lower whiskers, particularly in the "Stay" group. These outliers indicate that a subset of students intends to remain in the Czech Republic despite reporting low satisfaction levels (scores approaching 1.0) across these domains. This suggests that while positive experiences strongly encourage retention, negative experiences do not automatically result in departure, likely due to other overriding factors such as economic necessity or safety concerns in the home country.

Furthermore, the visible overlap between the distributions of the two groups confirms that while satisfaction is a differentiator, it is not a perfect predictor on its own, justifying the need for the multivariate logistic regression analysis that follows.

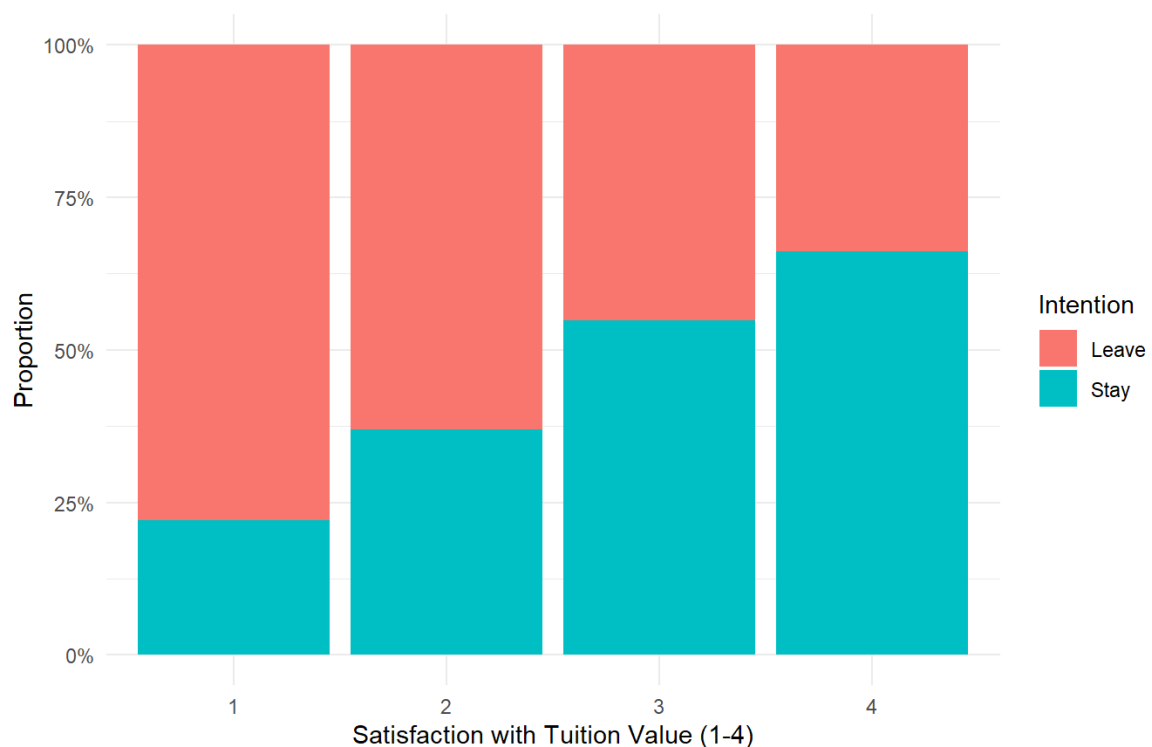
Single-Item Financial Factors

Following the analysis of the composite factors, I examined the single-item financial variables, beginning with Tuition. This variable measures how students rate the quality of their degree program relative to the tuition fees.

The descriptive statistics indicate a generally positive perception among respondents. The mean score is 3.35 (on the 1–4 scale), with a median of 3.00 and a 3rd quartile of 4.00. This suggests that most international students consider their education to be good value for money. To understand how this perception influences retention, I visualized the relationship in Figure 3.4.

Figure 3.19

Proportion of Students Wishing to Stay by Tuition Satisfaction (Source: Author's own processing using R)

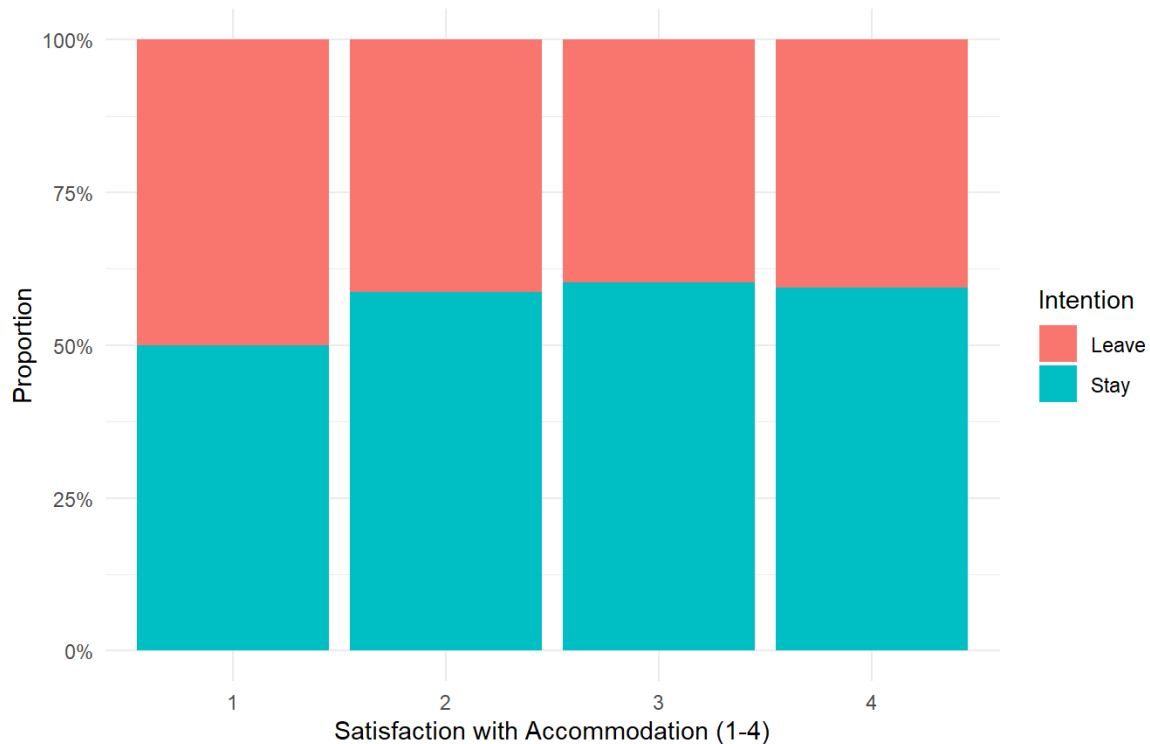


The chart reveals a distinct positive trend. As satisfaction with the value of tuition increases, the proportion of students intending to stay (represented by the turquoise bars) rises steadily. Among students who perceive the value as poor (score of 1), the intention to stay is low, at approximately 25%. However, this proportion grows with each step on the scale, reaching a clear majority (over 65%) among those who rated the value of their tuition highly (score of 4). This suggests that perceived economic value of education is a relevant factor in the decision to remain in the country.

Next, I analyzed the Accommodation variable, which measures satisfaction with housing costs. The descriptive statistics reveal a notably lower level of satisfaction compared to the other financial variables. The mean score is 2.61 (median 3.00), which is closer to the scale's midpoint than the high average seen for Tuition (3.35). This suggests that housing affordability is a more contentious issue for international students.

Figure 3.5

Proportion of Students Wishing to Stay by Accommodation Satisfaction (Source: Author's own processing using R)

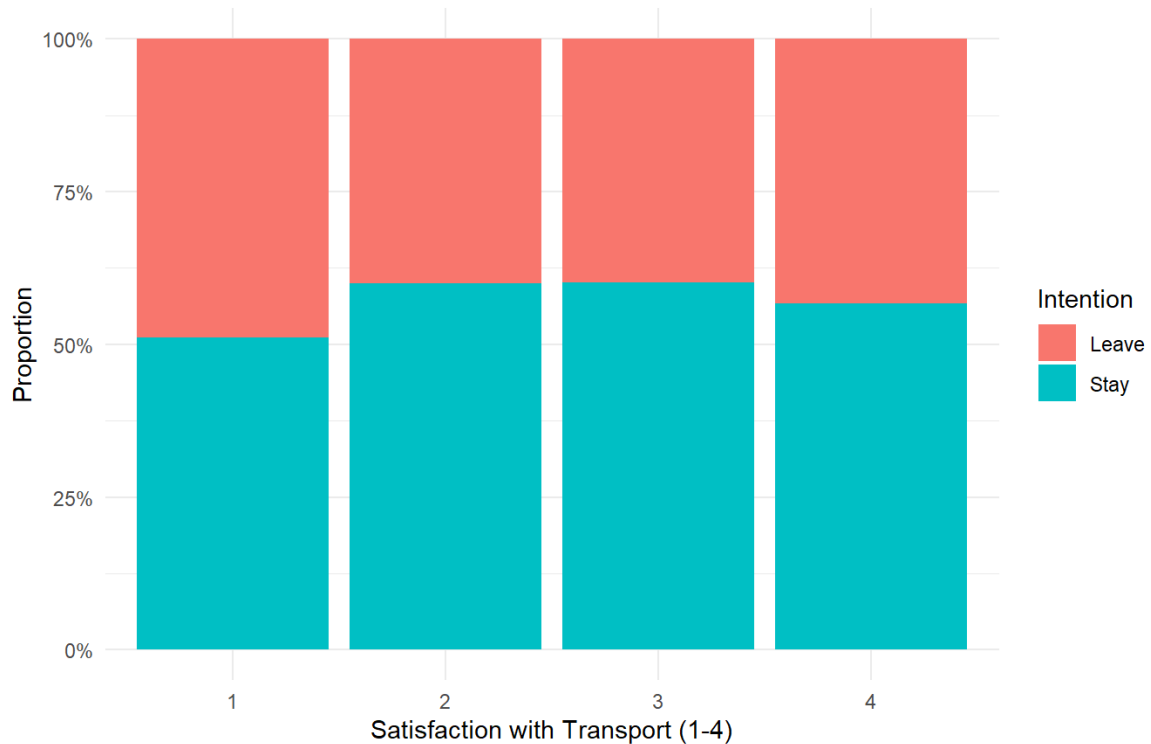


Unlike the clear linear trend observed with Tuition, this relationship appears to show a threshold effect. Students who are very dissatisfied with housing costs (score of 1) are split evenly, with only 50% intending to stay. However, for those who rate their satisfaction higher than 1, the proportion of students planning to stay stabilizes at approximately 60%, showing little variation across these higher levels. This suggests that while extreme dissatisfaction with housing costs may discourage retention, higher satisfaction does not necessarily drive it further, likely because housing is viewed as a basic necessity rather than a motivator.

I then examined the Transport variable, which assesses satisfaction with transportation costs. This variable shows the highest level of satisfaction among all financial indicators. The mean score is 3.49, and notably, the median is 4.00, meaning that at least half of the respondents gave this item the highest possible rating. This indicates that public transportation in the Czech Republic is widely perceived as affordable and accessible by international students.

Figure 3.6

Proportion of Students Wishing to Stay by Transport Satisfaction (Source: Author's own processing using R)

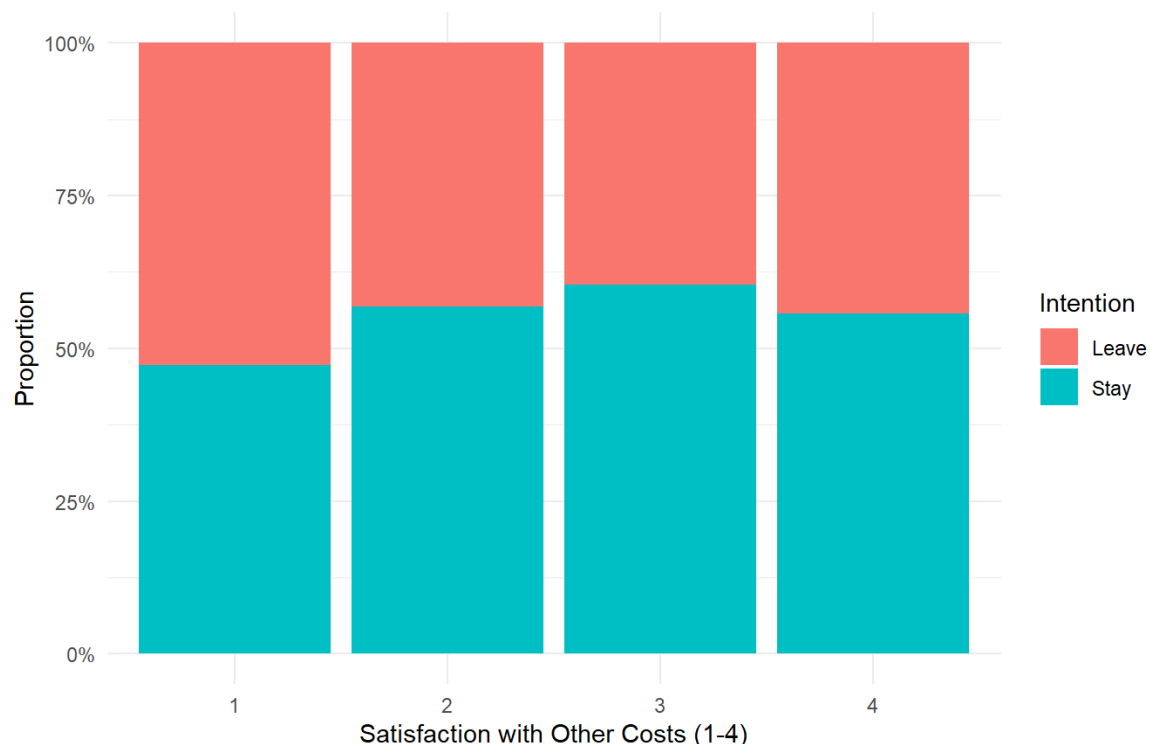


While satisfaction is generally high, the intention to stay does not increase linearly with higher scores. In fact, the proportion of students planning to stay is slightly lower for those who are most satisfied (score of 4) compared to those with moderate satisfaction (scores of 2 and 3). This "flat" or slightly declining trend suggests that while affordable transport is a positive feature of student life, it is likely a baseline expectation rather than a decisive driver for long-term retention. A student is unlikely to leave the country solely because of transport costs, but excellent transport alone is insufficient to convince them to stay.

Finally, I analyzed the Other Costs variable, which covers satisfaction with general living expenses such as food and daily necessities. The mean score is 2.86 (median 3.00), which places it between the highly rated Transport (3.49) and the lower-rated Accommodation (2.61). This indicates that while daily living costs are not as critical a pain point as housing, they are still a significant consideration for students.

Figure 3.7

Proportion of Students Wishing to Stay by Satisfaction with Other Costs (Source: Author's own processing using R)



The observed pattern is similar to the Transport variable. Students who are most dissatisfied with living costs (score 1) are the least likely to stay (under 50%). At moderate satisfaction levels (scores 2 and 3), the intention to stay rises to about 60%. Surprisingly, among the most satisfied students (score 4), this proportion does not increase and even drops slightly. This suggests that living costs are a 'hygiene factor': high costs drive students away, but low costs alone do not guarantee they will stay.

3.1.3 Independent Categorical Variables

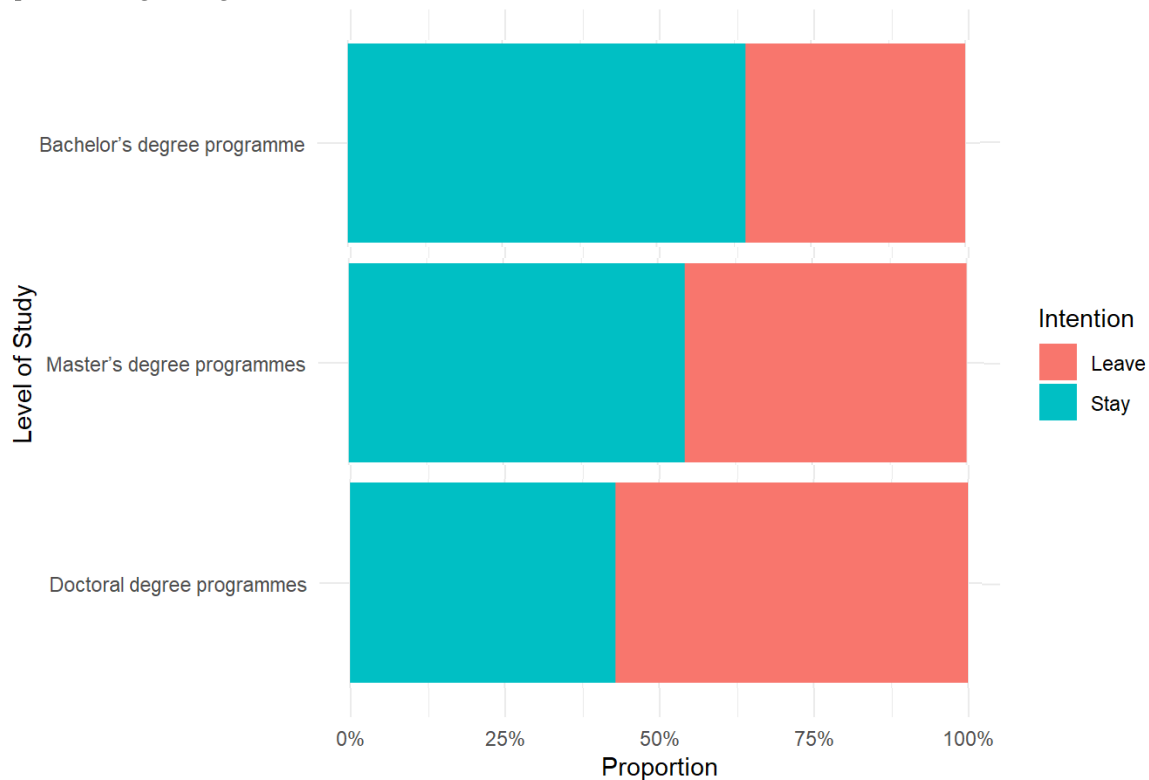
After analyzing the continuous factors, I focused on my categorical predictors: Level, Country, and Work. I again used the `summary(df_final)` output to understand their composition.

Level (Level of Study)

First, I examined the Level variable. The summary output showed the sample breakdown: Bachelor's degree programs (N=2,240), Master's degree programs (N=1,352), and Doctoral degree programs (N=633). To see if the intention to stay varied by level, I created the proportional bar chart in Figure 3.8.

Figure 3.8

Proportion of Students Wishing to Stay by Level of Study (Source: Author's own processing using R)



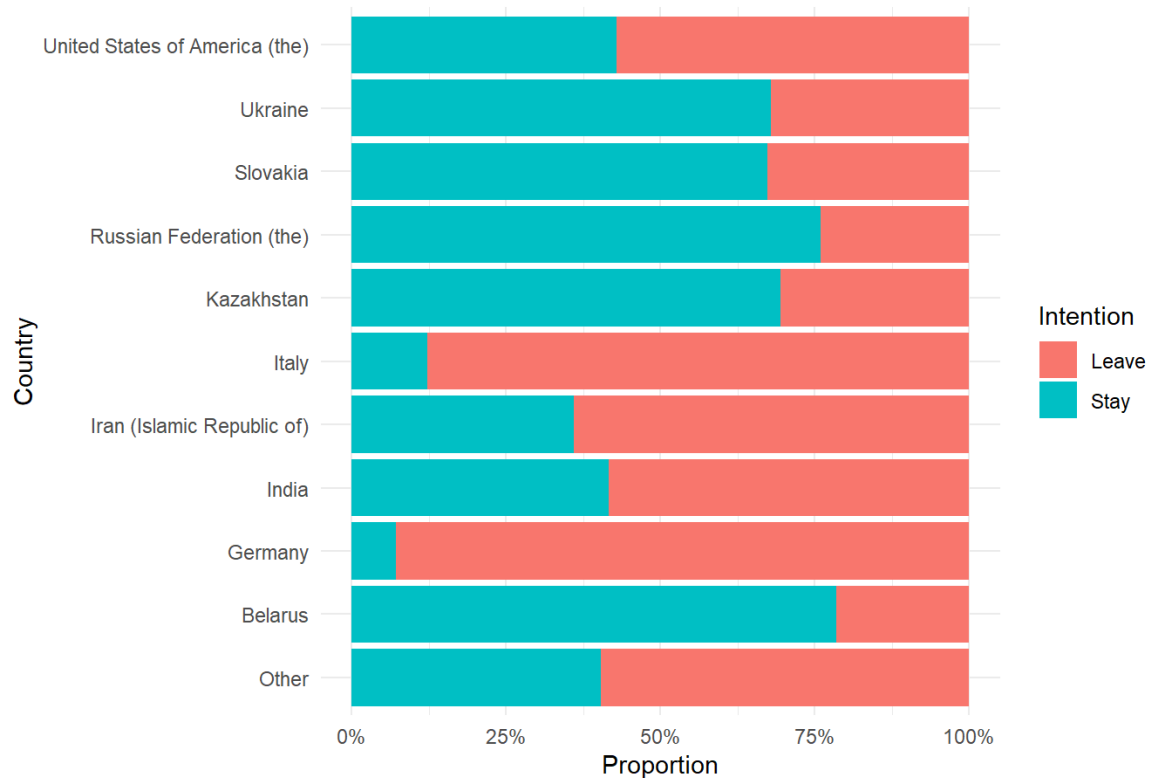
The chart revealed a clear inverse trend: the intention to stay decreases as the academic level increases. Bachelor's students showed the highest likelihood of staying (approximately 65%), while Master's students occupy a middle ground (approximately 55%). Notably, the trend reverses for the most advanced students; for the Doctoral group, a clear majority (approaching 60%) plan to leave the Czech Republic after graduation.

Country (Country of Origin)

Next, I analyzed the Country variable. The summary output showed the largest groups were from Slovakia (N=1,077), the Russian Federation (N=728), and Ukraine (N=492). The 'Other' category was also large (N=1,148). To explore this, I generated the chart in Figure 3.9.

Figure 3.9

Proportion of Students Wishing to Stay by Country of Origin (Source: Author's own processing using R)



This visualization shows that country of origin is a strong predictor. I saw sharp contrasts: students from Belarus, the Russian Federation, Kazakhstan, Ukraine, and Slovakia showed a high intention to stay (ranging from approximately 65% to 80%). Conversely, students from nations like Germany and Italy showed a strong intention to leave, with less than 15% planning to stay.

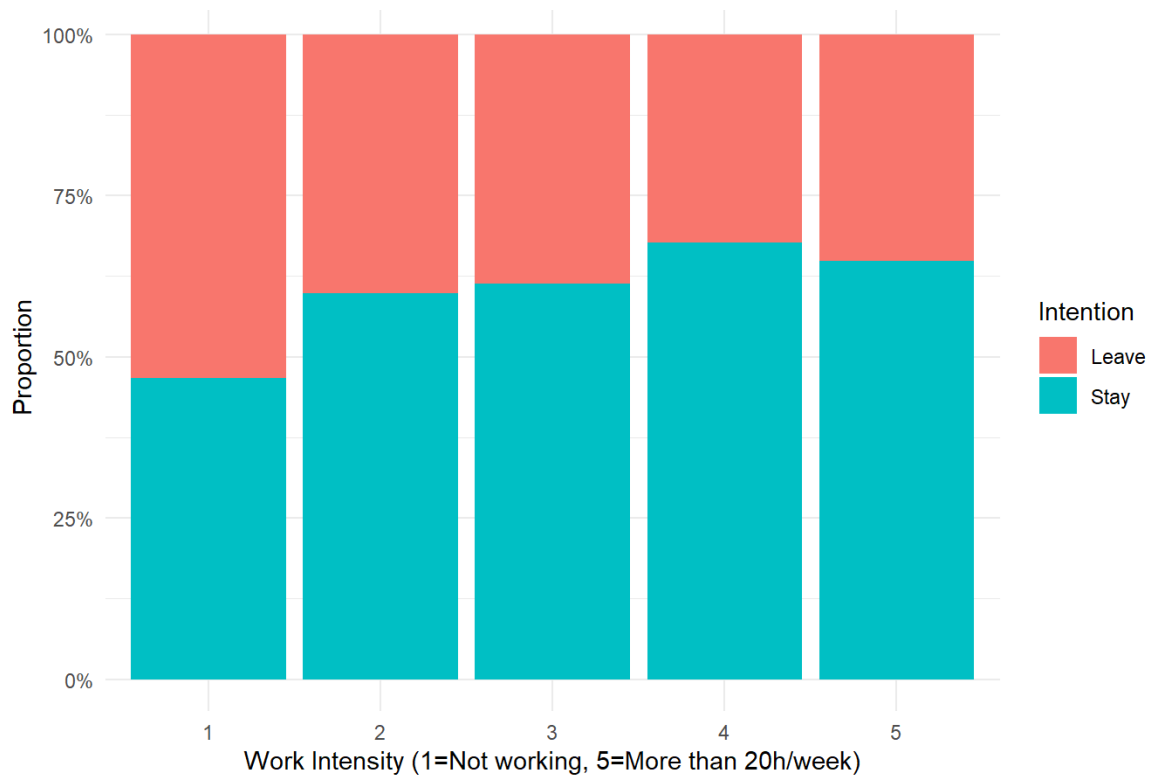
Work

Finally, I examined the Work variable. The mean score was 2.80 (on a 5-point scale), but the median (2.00) and 1st Quartile (1.00) were more revealing. This suggested that a large portion of the sample either does not work (score 1) or works only occasionally (up to 10 hours/week, score 2). I plotted this relationship in Figure 3.10.

This chart reveals a positive relationship. The more hours a student works, the more likely they generally are to plan to stay. Students who do not work (Category 1) were the least likely to stay (just under 50%). This proportion increases significantly for those who work, peaking with Category 4 (approximately 68%) before dipping slightly for those with the highest work intensity (Category 5), where approximately 65% of students intended to stay.

Figure 3.10

Calibration Plot (Source: Author's own processing using R) (Source: Author's own processing using R)



3.1.4 Note on Variable Selection: Exclusion of Gender

During the initial model-building phase, I considered Gender as a potential demographic predictor. To test its value, I ran a logistic regression model that included it with all my other predictors (as shown in Code 3.3)

Based on this analysis, I made the decision to exclude Gender from the final model. This decision was driven by the principle of statistical parsimony, supported by two key findings:

1. **Lack of Statistical Significance:** As shown in Code 3.3, the variable yielded a p-value of 0.076. More importantly, its inclusion led to a slight deterioration in the model's overall quality. The model including gender produced a Nagelkerke Pseudo- R^2 of 0.2155 (as shown in Code 3.3), which is lower than the 0.2225 achieved by the final model excluding it. This indicates that adding gender adds complexity without contributing to the model's explanatory power.
2. **Absence of Descriptive Difference:** To confirm the regression results, I performed a descriptive check by plotting the intention to stay by gender.

Code 3.659

Regression model with Gender variable (Source: Author's own processing using R)

```
## Call:
## glm(formula = Stay ~ ., family = "binomial", data = df_final)
...
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
##
## GenderMale          0.127176    0.071572   1.777  0.075585 .
...
## Null deviance: 5585.7  on 4115  degrees of freedom
## Residual deviance: 4868.1  on 4093  degrees of freedom
...
## N: 4116
## Nagelkerke R2: 0.2155
```

Figure 3.11

Proportion of Students Wishing to Stay by Gender (Source: Author's own processing using R)

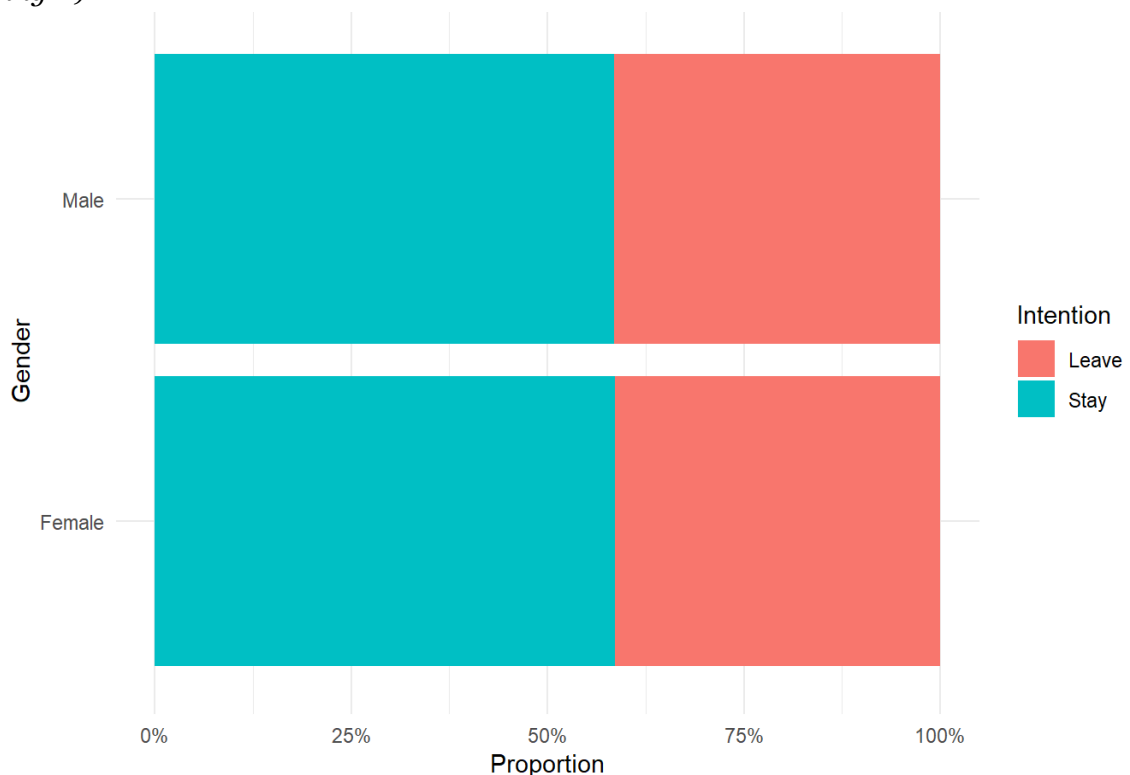


Figure 3.11 illustrates that the proportions are nearly identical. Female students and male students share almost the same likelihood of intending to stay.

Since the variable was neither statistically significant nor showed any real difference in the descriptive analysis, there was no reason to include it. Removing it also allowed me to keep

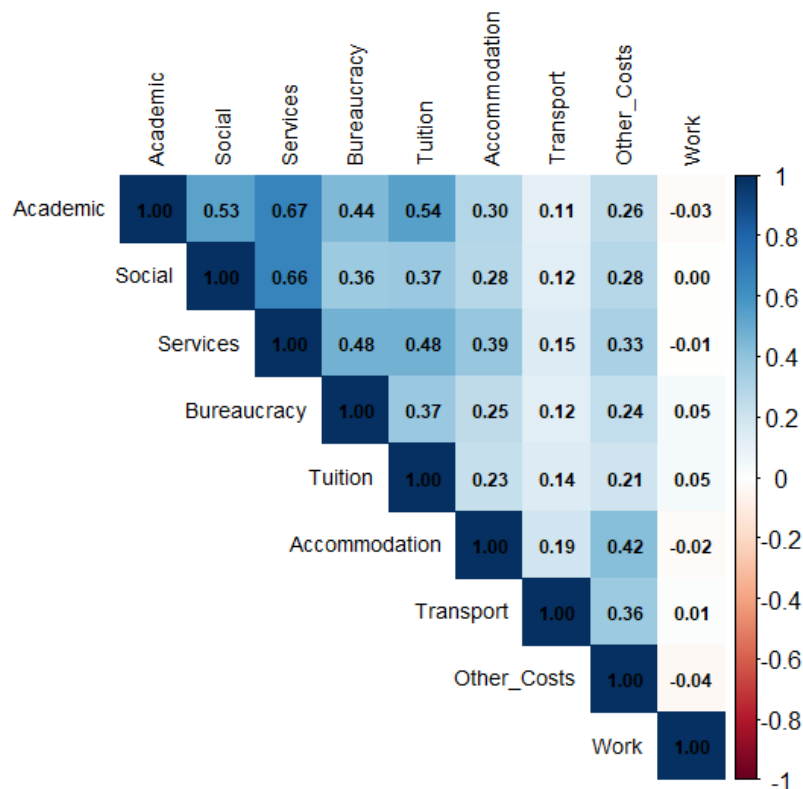
a fuller dataset (N=4,225) instead of losing data from students who did not specify their gender.

3.1.5 Multicollinearity check

Before proceeding to the main regression analysis, I performed a crucial assumption check for multicollinearity. This step was necessary to ensure my continuous predictor variables were not too highly correlated with each other. Strong multicollinearity could inflate the standard errors and make the model's final coefficient estimates unreliable.

To perform this check, I calculated and visualized a Pearson correlation matrix for all nine continuous predictors: the four composite factors (Academic, Social, Services, Bureaucracy), the four financial variables (Tuition, Accommodation, Transport, Other Costs), and the Work variable. I used the corrplot package in R to generate the chart in Figure 3.12.

Figure 3.12
Correlation Matrix Between Factors (Source: Author's own processing using R)



After generating the matrix, I examined the coefficients. The analysis confirmed that no problematic multicollinearity was present.

After generating the matrix, I examined the coefficients. The analysis confirmed that no problematic multicollinearity was present.

I observed that all correlation coefficients were at low to moderate levels. The highest values I found were 0.67 between the Services and Academic factors, and 0.66 between the Services and Social factors. While this indicates a moderate positive relationship (which is logical, as good services likely relate to good academic and social experiences), these values are still well below the commonly accepted critical threshold of 0.80. I also noted that the financial variables did not introduce high correlations; for instance, the highest correlation involving a financial metric was 0.54 (between Tuition and Academic).

Finally, the Work variable displayed negligible (near-zero) correlations with all other student experience factors. This is a crucial finding because it suggests that a student's decision to work probably is not driven by their happiness with the academic program or social life. This independence means that 'Work' contributes unique explanatory power to the model, capturing a specific aspect of retention that is not covered by the satisfaction measures.

Since I found no evidence of high multicollinearity that would compromise the model's validity, I concluded that all nine predictors were sufficiently independent. Therefore, I retained all predictors for the final logistic regression model.

3.2 Results of the Logistic Regression Model

After completing the descriptive analysis and assumption checks, I ran the binary logistic regression model, which forms the core of my analysis. This section presents the main findings.

As explained in the methodology, I chose to estimate a single, comprehensive model. This approach allowed me to assess the impact of the four composite factors (Academic, Social, Services, Bureaucracy) and the four single-item financial variables (Tuition, Accommodation, Transport, Other Costs) on the intention to stay, while simultaneously controlling for the strong influence of variables like work intensity, level of study, and country of origin.

To do this, I used the `glm()` (Generalized Linear Model) function in R, setting `family = "binomial"` as required for logistic regression. I regressed the dependent variable Stay against all predictor variables in the dataset. I then called the `summary()` function to get the detailed results, including coefficients, standard errors, and p-values for each predictor.

The first check was the overall model significance. I compared the Null deviance (5750.7) with the Residual deviance (4984.1). The large drop in deviance confirms that the comprehensive model provides a significantly better fit to the data than a null model with no predictors.

Code 3.859

Final regression model (Source: Author's own processing using R)

```
## Call:
## glm(formula = Stay ~ Academic + Social + Services + Bureaucracy +
##      Work + Level + Country + Tuition + Accommodation + Transport +
##      Other_Costs, family = "binomial", data = df_viz)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -3.51467    0.31803  -11.051  < 2e-16 ***
## Academic           0.28769    0.08670   3.318 0.000906 ***
## Social             0.22727    0.07412   3.066 0.002167 **
## Services           0.15322    0.09715   1.577 0.114768
## Bureaucracy        0.20184    0.08874   2.274 0.022940 *
## Work              0.18478    0.02178   8.483  < 2e-16 ***
## LevelDoctoral degree programmes -0.44513    0.11161  -3.988 6.66e-05 ***
## LevelMaster's degree programmes -0.08491    0.08129  -1.045 0.296229
## CountryBelarus     1.39672    0.24358   5.734 9.81e-09 ***
## CountryGermany    -1.93398    0.43662  -4.429 9.45e-06 ***
## CountryIndia        0.07092    0.16039   0.442 0.658363
## CountryIran (Islamic Republic of) 0.08570    0.31374   0.273 0.784729
## CountryItaly       -1.40398    0.39522  -3.552 0.000382 ***
## CountryKazakhstan  0.97526    0.17995   5.420 5.97e-08 ***
## CountryRussian Federation (the) 1.25290    0.11718  10.692  < 2e-16 ***
## CountrySlovakia    0.71721    0.10330   6.943 3.84e-12 ***
## CountryUkraine      0.79254    0.12330   6.428 1.29e-10 ***
## CountryUnited States of America (the) 0.11053    0.26214   0.422 0.673292
## Tuition            0.14018    0.05850   2.396 0.016568 *
## Accommodation      -0.01294    0.04026  -0.321 0.747879
## Transport          -0.07434    0.05322  -1.397 0.162471
## Other_Costs        -0.01127    0.05184  -0.217 0.827934
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 5750.7  on 4224  degrees of freedom
## Residual deviance: 4984.1  on 4203  degrees of freedom
## AIC: 5028.1
##
```

Model Explanatory Power

After confirming the model's statistical significance (shown by the large drop in deviance from 5750.7 to 4984.1), my next step was to quantify its overall explanatory power.

Standard R-squared (R^2) from linear regression is not applicable here, so a "pseudo-R-squared" metric is required. As noted in my methodology, I selected the Nagelkerke Pseudo R^2 . This metric is widely used because it scales from 0 to 1, making it easier to interpret in a way similar to standard R^2 .

I used the `NagelkerkeR2()` function from the `pscl` package to calculate this value. The function returned a Nagelkerke R^2 of 0.223. This result indicates that the combination of my predictors, the four composite factors, the four single-item financial variables, and the control variables (Work, Level and Country), explains approximately 22.3% of the variance in a student's intention to stay.

In social science, this is a moderate and meaningful level of explanatory power. I am modeling a complex human decision, which is influenced by many personal and economic factors not captured in this survey. Therefore, finding that my model accounts for over one-fifth of the variation confirms it is not just statistically significant but also practically relevant. An R-squared value in this range is acceptable in social science research, provided the explanatory variables are statistically significant (Ozili, 2022, p. 8).

Preparing for Interpretation: Odds Ratios

The next step was to interpret the effect of each individual predictor. The raw coefficients (Estimates) from the `summary()` output (in Code 3.4) are in units called log-odds, which are not very intuitive.

To make my results understandable, I converted these log-odds coefficients into Odds Ratios (OR). I did this by exponentiating the coefficients, using the `exp(coef(final_model))` command in R

Code 3.955

Odds Ratios for Predictors in the Final Model (Source: Author's own processing using R)

```
##          (Intercept)          Academic
##          0.02975775          1.33334894
##          Social          Services
##          1.25516891          1.16558644
##          Bureaucracy          Work
##          1.22365196          1.20294946
##          LevelDoctoral degree programmes          LevelMaster's degree programmes
##          0.64074126          0.91859637
##          CountryBelarus          CountryGermany
##          4.04191684          0.14457220
##          CountryIndia          CountryIran (Islamic Republic of)
##          1.07349429          1.08948222
##          CountryItaly          CountryKazakhstan
##          0.24561852          2.65186524
##          CountryRussian Federation (the)          CountrySlovakia
##          3.50047924          2.04871460
##          CountryUkraine          CountryUnited States of America (the)
##          2.20900810          1.11686902
##          Tuition          Accommodation
##          1.15047956          0.98714380
##          Transport          Other_Costs
##          0.92835455          0.98879483
```

This command provided a clear set of Odds Ratios (as shown in Code 3.5). As explained in the methodology, an OR greater than 1 signifies a positive influence on the intention to stay (it increases the odds), while an OR less than 1 signifies a negative influence (it decreases the odds). I used these Odds Ratios for my final interpretation in the following sections.

3.2.1 Interpretation of Significant Predictors

After calculating the Odds Ratios (ORs), I began interpreting each predictor's effect. I checked the p-value for each variable from the summary() output (Code 3.4) to determine its statistical significance (using $p < 0.05$ as the threshold). For significant variables, I used the OR to explain the direction and strength of their effect on the intention to stay.

I structured this interpretation by first examining the student experience factors, followed by the situational and demographic factors.

Student Experience Factors

I first analyzed the composite satisfaction factors and the financial variables. The model output showed that four of these variables were statistically significant predictors: three composite factors and one financial variable.

- **Academic Factor:** This emerged as a highly significant, positive predictor ($p < 0.001$, specifically 0.0009). Its OR was 1.333. This finding is clear: for every one-unit increase in a student's academic satisfaction score (on the 4-point scale), their odds of intending to stay increase by 33.3%, holding all other variables constant. This aligns with the logic that a high-quality education is a primary driver for students.
- **Social Factor:** Social integration was also a significant predictor ($p = 0.002$). The OR was 1.255. I interpret this as: a one-unit increase in the social integration score increases the students' odds of staying by 25.5%. This supports the hypothesis that feeling included in the university and local community is a key part of the decision.
- **Bureaucracy Factor:** Satisfaction with pre-arrival administrative procedures was also significant ($p = 0.023$). With an OR of 1.224, I concluded that a one-unit increase in this score is associated with a 22.4% increase in the odds of staying. This suggests a clear visa and admissions process has a positive impact on a student's desire to remain.
- **Tuition Satisfaction:** Among the financial variables, satisfaction with tuition value was the only significant predictor ($p = 0.017$). Its OR was 1.150. This indicates that for every one-unit increase in perceived value for money, the odds of staying increase by 15.0%. This highlights that while general living costs (accommodation, transport) may not determine retention, the specific economic value of the degree itself does.

Note on Non-Significant Factors

When I examined the summary() output, the Services factor ($p = 0.115$) and three of the financial variables (Accommodation ($p = 0.748$), Transport ($p = 0.162$), and Other Costs ($p = 0.828$)) were not statistically significant.

This suggests that once all other variables (especially demographics, work, academics, and tuition value) are accounted for, satisfaction with university facilities (like IT or canteens) and general living costs (housing, transport, daily expenses) do not have an independent, statistically reliable impact on the decision to stay. It appears that while high tuition value encourages retention, the broader cost of living is less decisive when other satisfaction factors are high.

Despite their lack of individual significance, I decided to retain these variables in the final model. I tested alternative models by removing them, but these reduced models demonstrated a poorer fit (indicated by a lower Nagelkerke R^2 and a higher AIC). Therefore, keeping them ensures the model remains theoretically comprehensive, controlling for the economic context even if specific costs are not primary drivers of the decision.

Situational and Demographic Factors

Next, I analyzed the situational and demographic variables. I found that these were, in fact, among the strongest and most important predictors in my entire model.

- **Work Intensity:** This variable was highly significant ($p < 0.001$) with an OR of 1.203. This provides a clear takeaway: for each one-unit increase on the 5-point work intensity scale (e.g., moving from "not working" to "working occasionally"), a student's odds of staying increase by 20.3%. This strongly suggests that integrating into the Czech labor market during studies is a powerful factor in the decision to stay.
- **Level of Study:** This variable showed a specific effect. I found no statistical difference between Master's students and the Bachelor's (reference) group ($p = .246$). However, the Doctoral degree variable was highly significant ($p < 0.001$) with an OR of 0.641. I interpreted this to mean that Doctoral students have 35.9% lower odds ($1.00 - 0.641$) of intending to stay compared to Bachelor's degree students.
- **Country of Origin:** This was the single most powerful predictor in the model. The results showed vast differences between nationalities, all relative to the "Other" reference category.
 - **Positive Influence:** Students from several Eastern European and Central Asian countries had a significantly higher likelihood of staying. The results were striking. The odds of staying were over four times higher for students from Belarus (OR = 4.04; $p < 0.001$) and nearly four times higher for students from the Russian Federation (OR = 3.50; $p < 0.001$). Students from Kazakhstan (OR = 2.65; $p < .001$), Ukraine (OR = 2.21; $p < 0.001$), and Slovakia (OR = 2.05; $p < 0.001$) all had more than double the odds of staying compared to the "Other" group.
 - **Negative Influence:** Conversely, students from some Western nations showed a significantly lower likelihood of staying. Students from Germany (OR = 0.15; $p < 0.001$) had their odds of staying reduced by 85% ($1.00 - 0.15$). Similarly, students from Italy (OR = 0.25; $p < 0.001$) had their odds reduced by 75% ($1.00 - 0.25$).
 - **No Significant Difference:** Finally, I noted that for students from India ($p = 0.658$), Iran ($p = 0.785$), and the United States ($p = 0.673$), there was no statistically significant difference compared to the "Other" reference group.

3.2.2 Model Quality and Reliability

After interpreting the model's coefficients, my final step was to assess its reliability. I had already established its explanatory power (Nagelkerke $R^2 = 0.223$), which is a moderate and meaningful amount for this type of research.

However, I also needed to assess two other critical aspects of its performance: its discriminatory power (how well it distinguishes "Stay" from "Leave") and its calibration (how accurate its probability predictions are). To do this, I performed two key diagnostic tests.

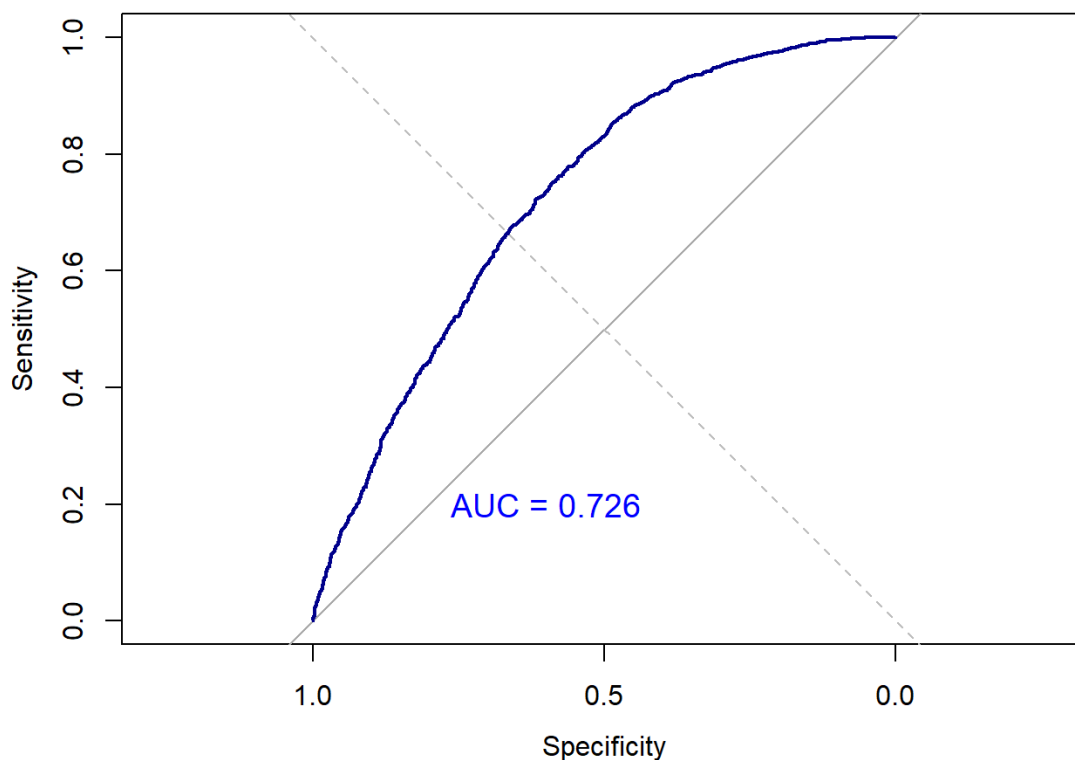
Discriminatory Power (AUC-ROC)

First, I measured the model's ability to discriminate between students who plan to stay and those who plan to leave. The standard method for this is the Receiver Operating Characteristic (ROC) curve and its corresponding Area Under the Curve (AUC) value.

To measure this, I used the `roc()` function from the `pROC` package. This function compares the true outcomes (`df_final$Stay`) against the predicted probabilities from my `final_model`. I then printed the AUC and plotted the ROC curve (Figure 3.13).

Figure 3.13

ROC Curve (Source: Author's own processing using R)



The code outputs a precise AUC value of 0.726. As noted in the methodology, an AUC of 0.5 (the gray line in the plot) signifies no discriminatory power (random chance), while an AUC of 1.0 would be perfect.

My result of 0.726 is generally considered to indicate "acceptable" or "good" discriminatory ability. The plot (Figure 3.13) visually confirms this, as the blue curve arches significantly away from the 0.5 line. I concluded that my model is reliably better than a random guess at distinguishing between students who intend to stay and those who plan to leave.

Calibration and Goodness-of-Fit

My second performance test assessed the model's calibration, or its reliability. I needed to know if its probability estimates are accurate. For instance, if the model predicts a 70% chance of staying for a group of students, I needed to check if, in reality, around 70% of those students actually intend to stay.

To evaluate this, I prepared the data. I combined the fitted(`final_model`) values (predicted probabilities) and the `final_model$y` values (actual 0 or 1 outcomes) into a `calibration_data` frame. Then, I used the `ntile()` function to divide the 4,225 students into 10 equal-sized groups (deciles) based on their predicted probability.

Finally, I created the `calibration_summary` table (Code 3.8). I grouped the data by these 10 deciles and calculated two metrics for each group:

- `mean_predicted`: The average predicted probability from the model.
- `act_prop`: The actual proportion of students in that group who intended to stay (the mean of the 0s and 1s).

Code 3.1031

Calibration Summary Table (Mean Predicted vs. Actual Proportion by Decile; Source: Author's own processing using R)

```
calibration_data <- data.frame(  
  predicted = preds,  
  actual = df_viz$Stay  
)  
  
calibration_data <- calibration_data %>%  
  mutate(  
    prob_decile = ntile(predicted, 10)  
  )  
  
calibration_summary <- calibration_data %>%  
  group_by(prob_decile) %>%  
  summarise(  
    mean_predicted = mean(predicted),  
    act_prop = mean(actual)  
  )
```

```
## # A tibble: 10 × 3
##   prob_decile mean_predicted act_prop
##   <int>      <dbl>      <dbl>
## 1       1      0.168      0.147
## 2       2      0.337      0.286
## 3       3      0.440      0.494
## 4       4      0.529      0.572
## 5       5      0.600      0.615
## 6       6      0.655      0.697
## 7       7      0.699      0.692
## 8       8      0.741      0.735
## 9       9      0.784      0.746
## 10      10      0.841      0.810
```

This table allowed me to directly compare the model's predictions to the actual outcomes. The results were very promising. For example, in the 8th decile, the model predicted an average probability of 74.1% (mean_predicted), while the actual observed proportion was 73.5% (act_prop). This close match indicates a high degree of calibration.

To visualize this agreement, I created the Calibration Plot (Figure 3.14), plotting the mean predicted values (x-axis) against the actual observed values (y-axis).

Figure 3.14

Calibration Plot (Source: Author's own processing using R)

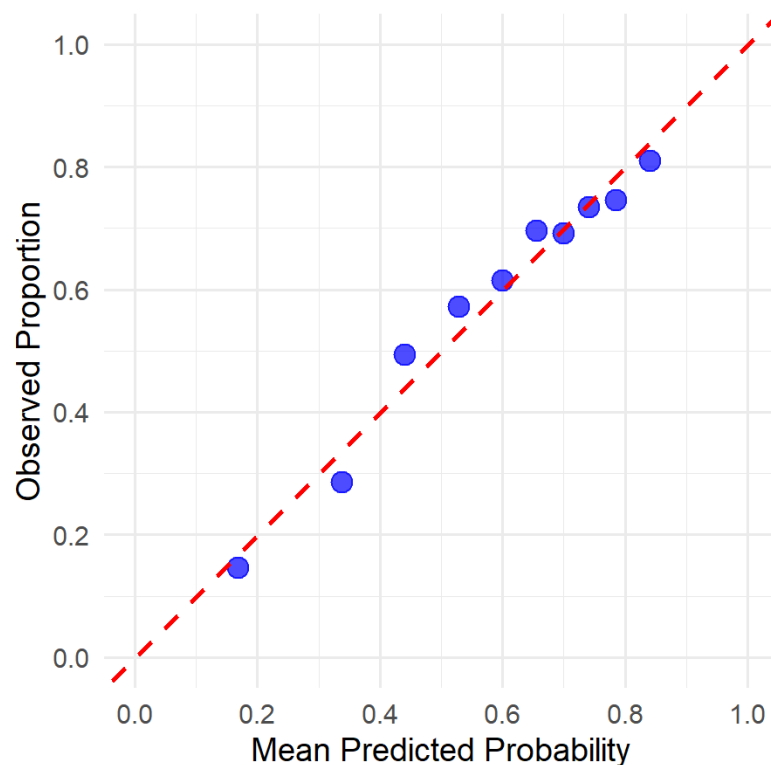


Figure 3.14 demonstrates a high degree of reliability. The 10 blue points lie very close to the ideal 45-degree (red dashed) line, which represents perfect calibration. This visual evidence,

supported by the data in the summary table, allowed me to conclude that my model is well-calibrated. Its probability estimates are reliable and closely align with the real-world outcomes in my sample.

Formal Goodness-of-Fit (Hosmer-Lemeshow Test)

After visually confirming the model's strong calibration, I performed one final, formal statistical test to assess its goodness-of-fit: the Hosmer-Lemeshow (H-L) test, a standard diagnostic for logistic regression.

I used the `hoslem.test()` function from the `pscl` package. This test is similar to my manual calibration plot and compares predicted probabilities to actual outcomes by calculating a Chi-squared statistic.

Code 3.1051

Hosmer and Lemeshow goodness of fit test (Source: Author's own processing using R)

```
# Hosmer and Lemeshow goodness of fit (GOF) test

# data: df_final$Stay, fitted(final_model)
# X-squared = 17.2, df = 8, p-value = 0.0281
```

The test returned a p-value of 0.0281 (Code 3.9). This presented an apparent contradiction. In an H-L test, a significant p-value ($p < 0.05$) typically suggests a poor model fit, indicating a significant difference between the model's predictions and the actual outcomes. This result conflicted with my excellent visual calibration plot (Figure 3.14).

I investigated this issue and found that the H-L test is known to be overly sensitive with large datasets. Since my sample is large ($N=4,225$), the test detects even tiny, unimportant differences and flags them as 'statistically significant,' even though they do not affect the model's reliability in practice.

This required a careful conclusion, weighing all the evidence:

- On one hand: My Nagelkerke R^2 (0.223) is moderate, and the H-L test is statistically significant ($p = 0.028$).
- On the other hand: My AUC (0.726) is good, and my visual calibration plot (Figure 3.14) shows excellent results.

Based on this, I concluded that my model is highly effective for its primary explanatory purpose. It successfully explains the relationships within my dataset. I can confidently stand by the findings that academic satisfaction, social integration, and country of origin are significant predictors because the model's ability to discriminate (AUC) and its probabilistic reliability (calibration plot) are both strong.

However, the significant H-L test and moderate R^2 suggest the model has limitations for predictive use. This is an expected outcome. I am modeling a complex human intention,

which is influenced by many personal and economic factors beyond the scope of a single survey.

In summary, the model provides a strong, robust, and reliable description of the factors influencing the 4,225 students in this sample; however, I must be cautious about using it to make precise predictions about future students.

4 Discussion of Key Findings

This chapter discusses the key factors identified in the statistical analysis that significantly influence the post-graduation intentions of international students in the Czech Republic. The goal is to compare the findings of this thesis with existing research and provide context for the observed relationships. While several students experience factors like academic satisfaction and social integration proved significant, demographic and situational variables, particularly the student's country of origin, emerged as exceptionally strong predictors of their intention to stay or leave after graduation.

4.1 Country of Origin

My analysis showed that country of origin was the most important factor in the model. Students from specific Eastern European and Central Asian countries were much more likely to plan on staying in The Czech Republic than the 'Other' nationalities group. This was especially true for students from Belarus (OR = 4.25), the Russian Federation (OR = 3.76), Kazakhstan (OR = 2.83), Ukraine (OR = 2.27), and Slovakia (OR = 2.19). In contrast, students from Germany (OR = 0.14) and Italy (OR = 0.22) were far less likely to plan on staying.

These findings are consistent with the DZS (2024) report, which used the same survey data. The report noted that while 48% of all international students plan to stay, this average hides major differences between groups (DZS, 2024, p. 146). Students from Russia and Central Asia often see staying as a "great opportunity," with many planning to work (39% and 36%) or continue studying (25% and 22%) in The Czech Republic (DZS, 2024, p. 146). This is a strong contrast to Western European students, as only 6% plan to work in The Czech Republic (DZS, 2024, p. 146). Western Europeans are also the most likely to leave, either to go home (31%) or to another country (39%), and the least likely to recommend studying in The Czech Republic. In comparison, very few students from Russia and Belarus (2%) plan to return home (DZS, 2024, pp. 111, 146–147).

This strong trend of students from former Soviet republics (CIS countries) staying, unlike Western Europeans, can be explained by the idea of competing migration subsystems (Ryazantsev et al., 2022). In this view, The Czech Republic receives migrants from the "overlapping area" (such as Belarus, Moldova, and Ukraine) between the traditional Eurasian (Russia-centered) system and the EU system. For CIS students, The Czech Republic can be a path to the EU job market that also feels culturally and linguistically familiar, especially for Slavic speakers (DZS, 2024, pp. 162–163). Cultural closeness is a known factor in drawing migrants in the post-Soviet region (Ryazantsev et al., 2022, p. 116). Other factors like established "migration corridors," history, and "push factors" (such as economic or political problems) at home may also make them more likely to stay (Ryazantsev et al., 2022, pp. 118, 123, 125). The DZS (2024) report confirms this, naming

safety as a key reason for students from Russia, Belarus, Kazakhstan, and Ukraine to choose The Czech Republic (p. 44).

In contrast, for students from Western EU countries like Germany and Italy, studying in The Czech Republic appears to be a temporary stay, fitting the concept of 'sojourn' described by Shen et al. (2022). Their home countries usually offer similar or better economic options, and easy, visa-free EU movement makes it simple to go home or move to another country (Ryazantsev et al., 2022, p. 117). The DZS (2024) report supports this, noting German students are less likely to recommend studying in The Czech Republic (p. 111) and may have higher expectations for university services based on standards at home (p. 165). This basic difference in goals and options likely explains why they are more likely to leave after graduation.

4.2 Work Intensity

Work intensity was the second strongest predictor: the more students worked, the more likely they were to stay in The Czech Republic. From the perspective of Human Capital Theory, this confirms that work is not merely a source of income but a mechanism for accumulating "country-specific human capital" (Calonge et al., 2023).

By working, students gain local experience, build networks, and understand the Czech job market. This makes staying more feasible and attractive, reducing uncertainty and increasing the opportunity cost of leaving. This early engagement smooths the student-to-work transition (Calonge et al., 2023).

Work also functions as a vital channel for social integration, facilitating the formation of "bridging social capital" that connects students to the host society beyond their immediate university bubble (Wang, 2020). The workplace helps students practice language (especially Czech), learn local work culture, and build social ties with colleagues. This complements academic connections and deepens their sense of belonging. While Lin (2025) notes that graduates can form 'parallel societies' if they struggle to integrate, successful integration through work seems to counteract this tendency. This enhanced social integration strongly reinforces the intention to remain in The Czech Republic.

4.3 Academic and Social Factors

My findings confirmed that higher academic satisfaction and greater social integration significantly correlated with students' intention to stay in The Czech Republic. This aligns with Kahu and Nelson's (2018) "educational interface" framework, suggesting that when institutional structures (curriculum, teaching) align with student needs, it fosters the engagement and self-efficacy necessary for retention. Quality education is a key motivator for study abroad, so academic satisfaction affects future plans. Social inclusion creates a sense of belonging, which is crucial for well-being.

Lu et al. (2024) emphasize that overall satisfaction stems from both academic quality and social interactions. However, they also note that satisfaction with local and domestic student interactions often ranks lower than academic satisfaction. This suggests that while social integration strongly predicts staying, achieving it remains a challenge for international students.

4.4 Bureaucratic Experience

Satisfaction with pre-arrival bureaucracy (admissions, visas, education recognition) also positively predicted the intention to stay. These findings support Lee's (1966) concept of "intervening obstacles." The significance of this factor confirms that administrative barriers act as a friction mechanism in the migration process; when these barriers are perceived as high, the "psychological cost" of staying increases.

This suggests initial administrative experiences have a lasting impact. A smooth process reduces stress and signals a welcoming environment, while early difficulties may create negative impression of a country as a whole. Lu et al. (2024) focus on post-arrival adjustment, stating that managing challenges reduces stress and promotes positive experiences. A positive initial bureaucracy experience could facilitate this later adjustment, lowering the intervening obstacles and boosting the overall satisfaction that encourages staying.

Conclusion

This thesis aimed to find out why international students decide to stay in or leave the Czech Republic after their studies. I used a survey of 4,169 students and a statistical method called logistic regression. The results showed that the two strongest factors are the student's country of origin and how much they worked during their studies. Other important factors included being happy with their academic program, feeling socially included, and having a smooth visa and admissions process before arriving. Feelings about university services or value for money were not statistically significant predictors.

These results show that practical factors are key. The strong link between working and staying suggests that getting a job helps students get local experience and build connections, making it easier to stay after graduating. In addition, the student's home country matters a lot. This suggests that "push and pull" factors like the economy or politics back home are very important. For example, students from CIS countries might see the Czech Republic as a long-term opportunity, while students from Western Europe might only plan a short stay.

This study also found some specific details. Doctoral students are less likely to stay than Bachelor's students are. The findings about bureaucracy show that a student's first impression from the visa process matters. These details show that there is no single solution for all students. We need to remember that students from different places have different reasons for staying or leaving.

This study has some limits. The data is from one point in time, so we cannot know if working causes students to stay or if students who want to stay are more likely to work. The data is all self-reported and the survey was voluntary, which could bias the results. In addition, the model predicts what students plan to do, not what they will actually do. The model is good at explaining the data we have but it might not be perfect for predicting what new students will do.

In conclusion, the choice to stay or leave is complex. This thesis showed it depends heavily on a student's home country, work experience, and their academic and social life. This is important for universities and the Czech government. To keep talented graduates, they should not only improve student life but also make administrative tasks easier and help students find jobs. By understanding that different students have different needs, the Czech Republic can better support international graduates who want to stay, which helps both the students and the country.

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